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March 9, 2023

CONTENT 1 Official electronic copy

US Food and Drug Administration
Center for Devices and Radiological Health
Document Mail Center – WO66-0609
10903 New Hampshire Avenue
Silver Spring, MD 20993-0002

Attn: Arti Streepy
RE: P130008/S090/A__ High AHI High BMI Panel Track BDD
Response to FDA Deficiency Letter dated January 17, 2023

Dear Ms. Streepy,

This submission is in response to the FDA deficiency letter dated January 17, 2023. The deficiencies noted by FDA are presented in **bold** text and are followed by Inspire's response in plain text in the attached electronic copy of this submission.

One electronic copy of this submission has been provided by Inspire Medical Systems.

This submission contains confidential and trade secret information and Inspire hereby requests that it be given the maximum protection provided by law.

If FDA has any questions or further information is required, please contact Sanjana Jaiswal at 763.392.0165 (cell 720-921-4435) or sanjanajaiswal@inspiresleep.com.

Sincerely,

A handwritten signature in blue ink, appearing to read "Terra Kissel", is written over a printed name and title.

Terra Kissel
Director, Regulatory Affairs

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RESPONSE TO LETTER DATED JANUARY 17TH 2023

Inspire is providing responses to the questions FDA provide in the letter dated January 17th, 2023

The original text from the letter is provided in **bold** text and Inspire's responses are in plain text. The sections within the letter and numbering is maintained as section headers to organize the response and for ease of navigation.

REAL WORLD DATA (RWD) AND REAL WORLD EVIDENCE (RWE)

FDA Letter Text

You have not provided adequate information to allow the assessment of the relevance and reliability of the ADHERE registry (the RWD) where RWE was gathered from to support the proposed indication expansion. For details on how FDA determines the suitability of RWD, please refer to the CDRH 2017 guidance [‘Use of Real-World Evidence to Support Regulatory Decision-Making for Medical Devices’](#)

FDA QUESTION 1 PART A

Generalizability: it is unclear if the patient population captured in the ADHERE registry is representative of the general population relying on this device.

Part A - Generalizability of Registry Population to Larger Device User Patient Population

In the Section MARKETING HISTORY (Page 15 of 60), you state that “Inspire therapy received FDA approval (P130008 incorporated herein by reference) in April 2014. As of October 11, 2022, there have been 31,185 implants of the Inspire system in the United States”. It is unclear the total number of the patients enrolled in the registry. Additionally, one of the two exclusion criteria for the ADHERE registry (protocol page 5 of 16) states that “Any reason the clinician deems patient is unfit for participation in the study” will be ineligible to participate in the registry. This is concerning because cases enrolled in the registry may not be representative, (i.e., an all-comer's design or consecutive patient enrollment). Please elaborate in what situation(s) a patient would be deemed as unfit for participating in the registry and how the current patient selection and enrollment criteria influence the potential for selection bias. This is to ensure that patients are naturally and routinely enrolled in the registry and that those who enrolled are a representative real-world user population.

INSPIRE RESPONSE TO QUESTION 1 PART A

The marketing history of the submission provides (31,185) total number of implants for Inspire therapy since its market release. The ADHERE Registry is an ongoing registry of patients implanted with the Inspire UAS system.

The registry includes centers who have interest and capability (research staff available) to collect and report clinical data related to Inspire Therapy. IRB approval is required for each clinical center to be

included in the Registry. In addition, subjects must consent to have their data collected and reported (unless a waiver of consent was provided by the IRB). Accordingly, not all implants are part of the registry.

A total of 3,988 subjects were enrolled in the ADHERE Registry as of July 22, 2022. **Figure 1** below is provided to further clarify the total ADHERE patient population, as well as the analysis population for both the AHI analysis and the BMI analysis included in this submission.

It should be noted that enrolled patients that did not receive the implant, had AHI<15, or did not have a baseline AHI recorded were excluded from the analysis included in this submission (total of 289 patients excluded for these reasons). As a result, 3,699 patients enrolled could be considered for this analysis at implant. However, of those 3,699 only 3,603 had baseline BMI as shown in **Figure 1**. As a result, a total of 96 patients did not have sufficient data available for the BMI analysis at implant. Post implant, patients were excluded from AHI and BMI analysis submitted due to missing final visit, no implant date or follow-up data, or final visit is beyond the analysis date (expected to occur in the future). This resulted in 1,542 patients included in the AHI analysis submitted and 1,508 included in the BMI analysis.

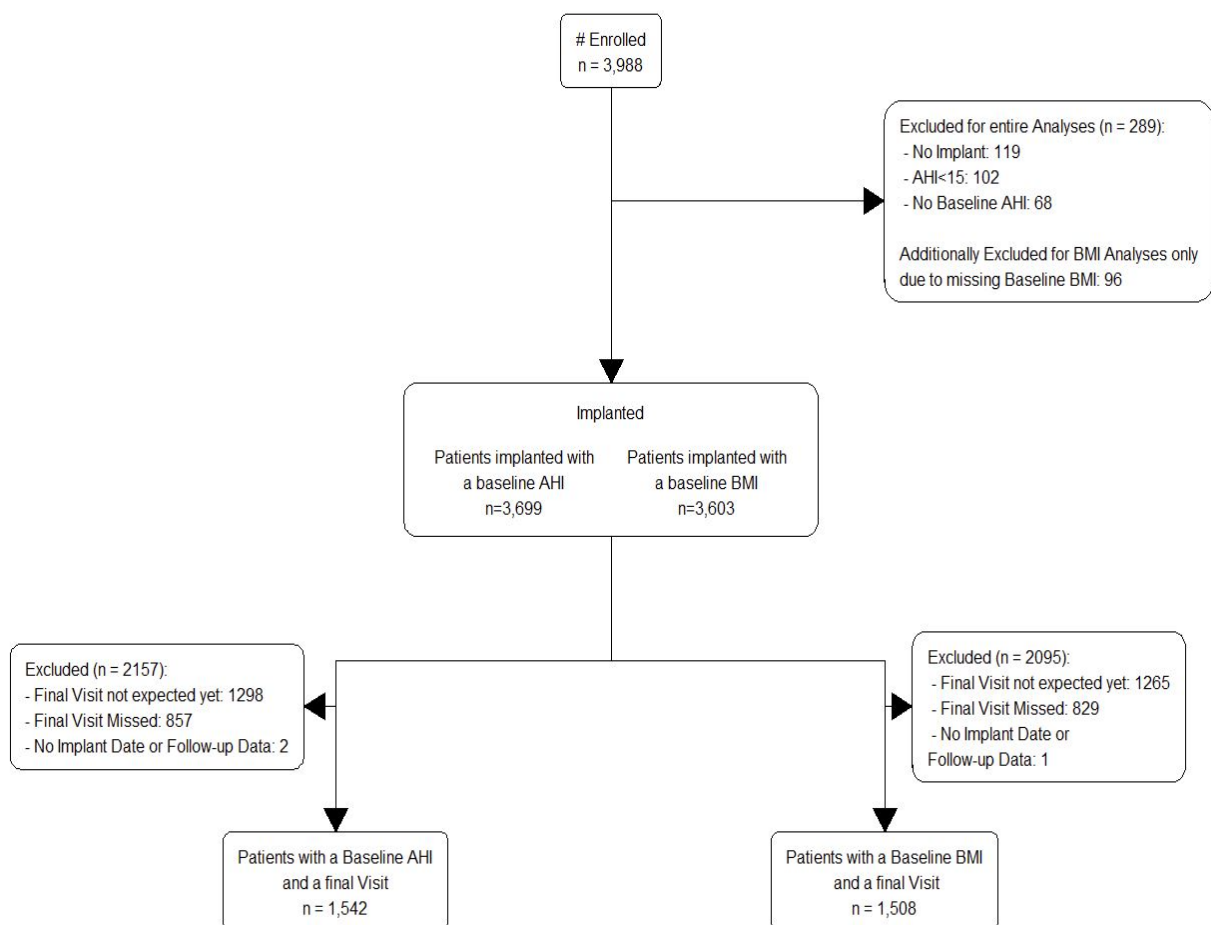


Figure 1. ADHERE Registry Patient Distribution

As further clarification related to the enrollment exclusion criteria (i.e., patient who is deemed unfit for participating in the registry), there are a few circumstances in which a patient may not be a good fit for the registry including:

- Patient was a referral from an outside clinic and will not be followed by the Registry center after implant of the Inspire system (will return to referral clinical for long-term follow-up);
- If the clinical centers are a sleep center and the sleep center does not have access to the patient's implant information;
- Patient lives out of state or travels out of state for long periods (snowbirds, etc.);
- Patient has history of inaccurate self-reporting (e.g., in those patients with patients with diseases affecting recall, cognition, or mental status).

This registry exclusion criteria is intended to ensure patients are willing and available for follow-up in the registry. Inspire does not believe this exclusion criteria introduces intentional bias.

FDA QUESTION 1 PART B 1)

Part B - Generalizability of Patients with Follow-up Visit to Overall Registry Population

In Section ADHERE REGISTRY (Page 24 of 60), you state that “1542 patients enrolled into the ADHERE registry (by July 22nd, 2022), had a final follow up visit with at least one of the following values recorded at the final visit: AHI, ESS, Therapy Usage, Questionnaire data”. The following is unclear from the information you provided:

1) how many patients have been enrolled by timeline in the ADHERE registry and

INSPIRE RESPONSE TO QUESTION 1 PART B 1)

As shown in [Figure 1](#) provided with Response to Part A, a total of 3,988 subjects were enrolled in the ADHERE Registry as of July 22, 2022.

Of the 3,988 patients enrolled in the registry only 1542 patients were included in the AHI analysis. These patients had both baseline AHI and final visit data available for analysis. All 1,542 of these patients were included in the AHI analysis.

In addition, of these 1542 patients, 1,508 patients had baseline BMI, baseline AHI, and final visit data available to be included in the BMI analysis.

The remaining patients that were enrolled but not included in the submission either missed their final visit, did not have an implant date or follow-up data, or are not far enough along in their follow-up to report final visit data.

FDA QUESTION 1 PART B 2)

if there are any differences between the 1542 patients analyzed for the indication expansion versus the rest of the patients in the registry.

Please provide additional details regarding the total number of patients enrolled in the registry and provide patient accountability data for the duration of intended registry follow-up. Please additionally provide justification on the representativeness and generalizability of the study sample by comparing patient demographic and clinical characteristics between patients analyzed at final follow-up versus patients eligible for final follow-up but not followed within the ADHERE registry. This is to help evaluate whether the analyzed study sample is generalizable to the larger registry population being evaluated.

INSPIRE RESPONSE TO QUESTION 1 PART B 2)

Baseline, follow up, and final visit data for 1542 patients was analyzed for this submission and included in the AHI analysis. Additionally, 1508 patients from the 1542 patients completing final visit with AHI baseline data also had baseline BMI data and were analyzed in the BMI analysis. The rest of the registry patients not included in the data analysis of this submission were excluded due to missing implant date, follow-up data, missed final visit, or have not completed the final AHI/BMI visit. For details regarding total number of patients enrolled in the ADHERE registry please refer to response to Question 1 Part A of this letter.

Generalizability of data in the registry is detailed in the AHI and BMI reports provided as an [Attachment 01](#) and [Attachment 02](#) respectively, also is summarized below:

In the Label_Expansion_AHI_Report ([Attachment 01](#)) Table 26 shows the differences in demographic and clinical characteristics between patients who completed a final follow-up visit compared to those who missed the final visit. 64.3% of patients in the AHI cohort completed their final visit. There was a significant difference in Baseline BMI, prior CPAP use, and gender between the two groups. Patients in the AHI cohort who completed the final visit had a significantly lower BMI. Additionally, the completed visit cohort had a significantly lower proportion of females and significantly higher proportion of patients with prior CPAP use.

In the Label_Expansion_BMI_Report_ ([Attachment 02](#)) Table 26 shows the differences in demographic and clinical characteristics between patients who completed a final follow-up visit and those who missed the final visit. 64.5% of patients in the BMI cohort completed the final visit. There was a significant difference in baseline BMI and gender between the two groups. Patients in the BMI cohort who completed the final visit had a significantly lower BMI and a significantly lower proportion of females, as compared to the patients who missed the final visit. Final follow-up rates for patients in the BMI≤32, 32<BMI≤35, 35<BMI≤40, and BMI>40 groups were 66%, 60.7%, 52.8%, and 61.1%, respectively.

While there were some small differences between patients who completed their final visit compared to those who did not, Inspire concludes that the data is generalizable for the purpose of expanding the AHI indication and revising the BMI warning in Inspire's current labeling.

FDA QUESTION 2

Data Quality: In the submission, you provided the evaluation of the ADHERE registry data to support the indication expansion by indicating that the Inspire UAS system is safe and effective in treating OSA patients with AHI greater than 15 and less than or equal to 100. The data also is being utilized to support an update to the BMI warning to indicate that the therapy has been studied in higher BMI categories, with safe and effective use in patients with BMI less than or equal to 40.

However, you did not provide information that allows us to assess data quality of the ADHERE registry and whether measures are taken to ensure data quality and integrity. Please provide details regarding what measure(s) are taken to ensure the data quality of the ADHERE registry.

Characteristics of RWD data quality may include, but are not limited to,

- 1) data accuracy (e.g., outcomes are appropriately adjudicated and verified; data quality assurance procedure and data quality audit program that may help improve data quality) and
- 2) data completeness and comprehensiveness (e.g., relevant outcomes and patient population's experience are captured; missingness or out of range values are minimized and data elements are collected to adjust for confounding factors). Confidence in RWD data quality is essential to help determine reliability of the ADHERE registry as an RWE source. Please refer to Section V Characteristics of RWD in the [CDRH 2017 guidance](#) for more details on the quality of RWD.

INSPIRE RESPONSE TO QUESTION 2

To ensure the data quality/integrity of the ADHERE registry, the following measures have been taken:

- The ADHERE Registry includes clinical centers that have been trained on the registry protocol, as well as the clinical database (including entry of the clinical data).
- Clinical center personnel are not provided access to the clinical database until the training has been completed.
- Data is entered into a 21 CFR Part 11 compliant electronic data capture system, which includes role-based, secure access to the data.
- Data entry guidelines are provided to the clinical center in order to ensure accurate data collection and reporting. Edit checks are coded into the database to provide a foundation for clean data. Automatic edit checks are used in order to avoid illogical and out of range data. In addition, the data manager reviews data via reports in order to look for outliers and illogical data. Sites are contacted for any issues that need to be addressed outside of the automatic edit checks.

FDA QUESTION 3

Data Discrepancy: You provided the total number of subjects whose outcomes were evaluated and the number of subjects at baseline. However, these numbers are different across various tables and sections. In the HIGH BMI≤40 ADHERE DATA ANALYSIS (page 28 of 60), you state that 1508 patients enrolled into the ADHERE registry (by July 22, 2022) had BMI collected at baseline and had a final follow up visit with at least one of the following values recorded at the final visit: AHI, ESS, Therapy Usage, CGI, and/or Patient Satisfaction data. However, in the ADVERSE EVENTS section (page 31 of 60), the number of subjects in the Table 19. Implant Adverse Events by Seriousness and Baseline BMI is 3603. A similar discrepancy is also observed for the analysis by AHI. In ADHERE REGISTRY section (page 24 of 60), you state that 1542 patients enrolled into the ADHERE registry (by July 22nd, 2022), had a final follow up visit with at least one of the following values recorded at the final visit: AHI, ESS, Therapy Usage, Questionnaire data. However, in Attachment 3 (page 3) of your submission, the number of subjects in the Table 1: Implant Adverse Events by Seriousness and Baseline AHI is 3699. Please clarify why the total number of subjects whose outcomes are evaluated and number of subjects at baseline analyzed are different. This is to ensure transparency and eliminate confusion over study population included in data analysis.

INSPIRE RESPONSE TO QUESTION 3

The ADHERE Registry is a real-world ongoing registry. Due to the nature of the registry, subjects are in various states of follow-up. [Figure 1](#) provided in response to Part A of response #1 provides details regarding the total number of subjects enrolled in ADHERE, as well as the number of subjects that are included in either the AHI analysis or BMI analysis.

The total number of subjects enrolled in the ADHERE registry as of July 22, 2022 (date of data cut for analysis) was 3,988. This includes any subject that has been added to the database, even if only baseline data was reported for that subject at the time of the data analysis.

The subjects included in the analysis in Table 19. Implant Adverse Events by Seriousness and Baseline BMI were those who were enrolled in the ADHERE registry as of July 22, 2022, had a baseline BMI reported, and had data entered through implant, see [Figure 1](#) of this response. Patients who had not yet completed their final follow-up visit were included in the implant adverse event analysis (3603 total patients), while patients who had completed a final visit were included in the 1508 patients noted in HIGH BMI ≤40 ADHERE DATA ANALYSIS (page 28 of 60). The implant adverse and event reporting is specific to the implant time point and, thus, all patients with a Baseline BMI data entered through implant were included in the implant AE analysis to ensure all relevant safety information was reported.

Similarly, the subjects included in Table 11. Implant Adverse Events by Seriousness and Baseline AHI were those who were enrolled in the ADHERE registry as of July 22, 2022, had a baseline AHI reported, and had data entered through implant (3699 total patients), while the 1542 patients noted in the ADHERE REGISTRY section (page 24 of 60) included those who had also completed a final visit.

For reference, the implant adverse event analysis for the smaller cohorts of patients who had a baseline AHI/BMI, were implanted with Inspire UAS system, and completed a final visit is now included in Tables 15-22 in Label_Expansion_AE_Report_2023-02-14 ([Attachment 03](#)).

As described above, the total number of subjects included in the AHI analysis was 1,542. Of those 1,542 subjects, 1,508 had a baseline BMI and were therefore also included in the BMI analysis. In order to clarify the number of subjects in each analysis (and the number that contributed to each variable), a table which lists the N's associated with each analysis and variable is provided below.

Variable	AHI Analysis N	BMI Analysis ¹ N
Total N	1542	1508
Baseline AHI	1542	1508
Final AHI	1179	1154
Baseline BMI	1508	1508
Baseline ESS	1352	1326
Final ESS	1277	1256
Final Therapy Use	1189	1170
Reviewed for Implant Adverse Event	3699	3603
Reviewed for Follow-up Adverse Event	1542	1508
¹ All subjects in the BMI analysis group are also part of the AHI analysis group		

STATISTICAL ANALYSIS

FDA QUESTION 4

You have used the term “Sher Responder Rate” in the effectiveness analysis. However, you did not clearly define this term in the submission. To facilitate the agency’s interpretation of effectiveness data, we recommend that you provide a clear definition for the term.

INSPIRE RESPONSE TO QUESTION 4

The Sher responder rate is defined as the percentage of patients that have a greater or equal to 50% reduction in AHI and less than or equal to 20 events/hour¹.

¹ Sher AE, Schechtman KB, Piccirillo JF. The efficacy of surgical modifications of the upper airway in adults with obstructive sleep apnea syndrome. Sleep. 1996 Feb;19(2):156-77. doi: 10.1093/sleep/19.2.156. PMID: 8855039.

This responder rate is the most commonly used method of measuring success for surgical treatments of obstructive sleep apnea, used in multiple sleep apnea surgery trials, and was the primary endpoint in the Inspire PMA STAR trial².

In addition to these reasons, Inspire believes this responder rate is relevant to the high AHI and BMI >32 patient group who are CPAP intolerant for the following reasons:

- While oral appliances can sometimes be successful in mild to moderate OSA, oral appliance success rates are typically less than 50% effective in patients with severe OSA, such as AHI > 30.^{3, 4, 5, 6}
- Treatment for these patients with traditional irreversible, resective airway surgery typically has a response rate of 50% or less.^{7, 8}
- Higher BMI and AHI patients have been shown to be at higher risks for complications after traditional airway sleep surgery⁹, which may limit the access of these options for high BMI patients.
- Multiple studies have shown that UAS implant has a lower risk for surgical and post-operative complications and readmissions than traditional sleep surgeries⁶. (Nord, Oto Open 2022; Van Daele, Oto Open 2021)
- Additionally, UAS is an adjustable and reversible procedure unlike traditional airway surgery.

Therefore, Inspire feels that the defined responder rate is appropriate to demonstrate that UAS is a more effective and safer alternative than current options, including oral appliances and traditional sleep surgery, for higher AHI /BMI patients who are CPAP intolerant.

² Strollo PJ Jr, Soose RJ, Maurer JT, et al. Upper-airway stimulation for obstructive sleep apnea. *N Engl J Med*. 2014;370(2):139-149. doi:10.1056/NEJMoa1308659

³ Ferguson KA, Ono T, Lowe AA, Keenan SP, Fleetham JA. A randomized crossover study of an oral appliance vs nasal-continuous positive airway pressure in the treatment of mild-moderate obstructive sleep apnea. *Chest*. 1996;109(5):1269-1275.

⁴ Johnston CD, Gleadhill IC, Cinnamon MJ, Gabbey J, Burden DJ. Mandibular advancement appliances and obstructive sleep apnoea: a randomized clinical trial. *Eur J Orthod*. 2002;24(3):251-262.

⁵ Marklund M, Stenlund H, Franklin KA. Mandibular advancement devices in 630 men and women with obstructive sleep apnea and snoring: tolerability and predictors of treatment success. *Chest*. 2004;125(4):1270-1278.

⁶ Marklund M, Braem MJA, Verbraecken J. Update on oral appliance therapy. *Eur Respir Rev*. 2019;28(153):190083. Published 2019 Sep 25.

⁷ Aaron E. Sher, Kenneth B. Schechtman, Jay F. Piccirillo, The Efficacy of Surgical Modifications of the Upper Airway in Adults With Obstructive Sleep Apnea Syndrome, *SLEEP*, Volume 19, Issue 2, March 1996, Pages 156–177, <https://doi.org/10.1093/sleep/19.2.156>

⁸ Orosco RK, Lin HW, Bhattacharyya N. Safety of Adult Ambulatory Direct Laryngoscopy: Revisits and Complications. *JAMA Otolaryngol Head Neck Surg*. 2015;141(8):685–689. doi:10.1001/jamaoto.2015.1172

⁹ Kezirian EJ, Weaver EM, Yueh B, Khuri SF, Daley J, Henderson WG. Risk factors for serious complication after uvulopalatopharyngoplasty. *Arch Otolaryngol Head Neck Surg*. 2006 Oct;132(10):1091-8. doi: 10.1001/archotol.132.10.1091. PMID: 17043257.

FDA QUESTION 5

For BMI indication expansion, the stratum of BMI>40 was separately reported for safety analysis. However, it appeared that for the effectiveness endpoint analysis, the stratum BMI>40 was combined with the stratum 35<BMI≤40 into a larger stratum BMI>35. If the intention is to expand device labeling to include subjects with BMI≤40, then please report effectiveness endpoint separately on these two strata 35<BMI≤40 and BMI>40, to facilitate the agency’s evaluation of device effectiveness on the indicated population.

INSPIRE RESPONSE TO QUESTION 5

Additional analysis related to the two strata (35<BMI≤40 and BMI > 40) has been completed and is provided as [Attachment 02](#).

Tables 27-29 in the Label_Expansion_BMI_Report_Update report ([Attachment 02](#)) show baseline characteristics and final follow-up outcomes for the four BMI groups (BMI≤32, 32<BMI≤35, 35<BMI≤40, and BMI>40).

Per Table 27, at least one of the BMI groups was found to have a significantly different baseline AHI, with the higher BMI groups having a higher baseline AHI on average. These findings are similar to the findings previously reported for the three BMI groups (BMI≤32, 32<BMI≤35, and BMI>35). There are no other significant differences in baseline characteristics between the four BMI groups (apart from baseline BMI, which is expected).

Per Table 28 and 29, there is no significant difference between the four BMI groups in terms of final AHI, change in AHI (baseline to final), final Sher response rate, final ESS, change in ESS (baseline to final), and final Clinical Global Impression – Improvement Scale (CGI-I). However, at least one of the BMI groups is significantly different in terms of final therapy usage (hours/night), with the BMI>40 group having the lowest therapy use on average. Again, these findings are similar to the findings previously reported for the three BMI groups (BMI≤32, 32<BMI≤35, and BMI>35).

It should be noted that all groups’ therapy usage exceeded 4 hours per night over 5 nights per week which is the threshold for CPAP compliance. Additionally, for all groups, the Sher Response rate was at least 50% (BMI>40) and up to 65% (BMI≤32).

FDA QUESTION 6

On page 24 of the provided Executive Summary, you stated that ADHERE registry “capture outcomes of 5,000 patients implanted with the Inspire™ UAS System”. Then on page 33, you stated that “[t]here were 588 patients OUS (EU) and 954 patients in US centers within the ADHERE registry.” You also report the corresponding Pre-Enrollment data in Table 21. Please clarify on which subset of the registry data the Pre-Enrollment results are presented in Table 21.

INSPIRE RESPONSE TO QUESTION 6

ADHERE is an ongoing registry and we anticipate enrolling up to 5,000 patients. Table 21 provides a breakdown of the 1542 patients included in the analysis (as described in response #1 above) into US and European subjects.

FDA QUESTION 7

Per page 33 of the provided Executive Summary, ADHERE registry has enrollment from both US and OUS sites. To help the Agency's understanding of the generalizability of the observed device effectiveness performance on the proposed US indicated populations, please report effectiveness endpoint data for subjects with $65 < \text{AHI} \leq 100$ and for subjects with $32 < \text{BMI} \leq 40$, stratified by region (US vs OUS).

INSPIRE RESPONSE TO QUESTION 7

Additional analysis has been completed on the US vs OUS populations. The reports that describe this analysis are provided as [Attachment 01](#) and [Attachment 02](#).

Table 24 in the Label_Expansion_AHI_Report ([Attachment 01](#)) shows the differences in outcomes for patients with $65 < \text{AHI} \leq 100$ between US and OUS sites. There is no significant difference between these regions regarding change in AHI (Baseline – Final), final Sher response rate, final ESS, and final therapy usage. However, there is a significant difference in final AHI and change in ESS between the two regions. Final AHI is significantly lower and ESS reduction is significantly greater for patients implanted in the US compared to OUS patients.

Table 30 in the Label_Expansion_BMI_Report_Update ([Attachment 02](#)) shows the differences in outcomes for patients with $32 < \text{BMI} \leq 40$ between US and OUS sites. There is no significant difference between regions regarding final AHI, change in AHI (Baseline – Final), final Sher response rate, and change in ESS (Baseline – Final). However, there is a significant difference in final ESS and final therapy usage between the regions. Final ESS is significantly lower and final therapy usage (hours/night) is significantly higher in US patients as compared to OUS patients.

FDA QUESTION 8

You reported effectiveness data stratified by high enrolling site or not for subjects with $\text{AHI} > 65$ (Table 22) and for subjects with $\text{BMI} > 32$ (Table 23). This is not sufficient for the evaluation of heterogeneity of device effectiveness performance across individual study sites, which impacts the generalizability of the observed results. Please report effectiveness endpoint data for subjects with $65 < \text{AHI} \leq 100$ and for subjects with $32 < \text{BMI} \leq 40$, stratified by individual enrolling site.

INSPIRE RESPONSE TO QUESTION 8

Additional information related to outcomes by enrolling center are provided in [Attachment 01](#) and [Attachment 02](#).

Table 25 in the Label_Expansion_AHI_Report ([Attachment 01](#)) shows the outcomes for patients with $65 < \text{AHI} \leq 100$ stratified by site. No significance testing could be done on these outcomes as the sample sizes at each site were very small.

Table 31 in the Label_Expansion_BMI_Report([Attachment 02](#)) Update shows the outcomes for patients with $32 < \text{BMI} \leq 40$ stratified by site. Again, no significance testing could be done on these outcomes due to the small sample sizes at each site.



Label Expansion - AHI > 65

ADHERE Registry

Anna Menzl

02/02/23

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Demographics and Baseline Information

Table 1: Demographic Variables by Baseline AHI

Variable	AHI $\leq$ 65	AHI $>$ 65	p-value
N	1483 (96.2%)	59 (3.8%)	
Age	60.14 $\pm$ 10.62 (61), 21 - 85, N=1476	60.14 $\pm$ 9.93 (61), 37 - 78, N=59	0.996
AHI - Baseline	34.58 $\pm$ 12.62 (32.5), 15 - 65, N=1483	75.79 $\pm$ 9.97 (73), 65.64 - 118.7, N=59	0.000
BMI - Baseline	28.99 $\pm$ 3.76 (28.98), 17.71 - 47.19, N=1450	30.64 $\pm$ 3.55 (30.75), 21.8 - 38.35, N=58	0.001
ESS - Baseline	11.36 $\pm$ 5.55 (11), 0 - 24, N=1298	10.67 $\pm$ 5.45 (10.5), 0 - 22, N=54	0.365
Female	24% (355)	19% (11)	0.435
White	95.8% (1407)	93.2% (55)	0.319
Hispanic or Latino	1.9% (28)	1.7% (1)	1.000
Prior Oral Appliance	23% (339)	17.2% (10)	0.343
Prior CPAP	97.8% (1443)	100% (58)	0.631

Note:

Format for numeric variables: Mean $\pm$ SD (Median) Range

1542 patients enrolled into the ADHERE registry by July 22nd, 2022 had a final follow up visits with at least one of the following values recorded at the final visit: AHI, ESS, Therapy Usage, CGI, Patient Satisfaction (1483 with AHI $\leq$ 65 and 59 with a Baseline AHI $>$ 65). Patients with higher AHI have a significantly higher BMI at Baseline (p value=0.0011)

Outcomes at Final Visit

AHI

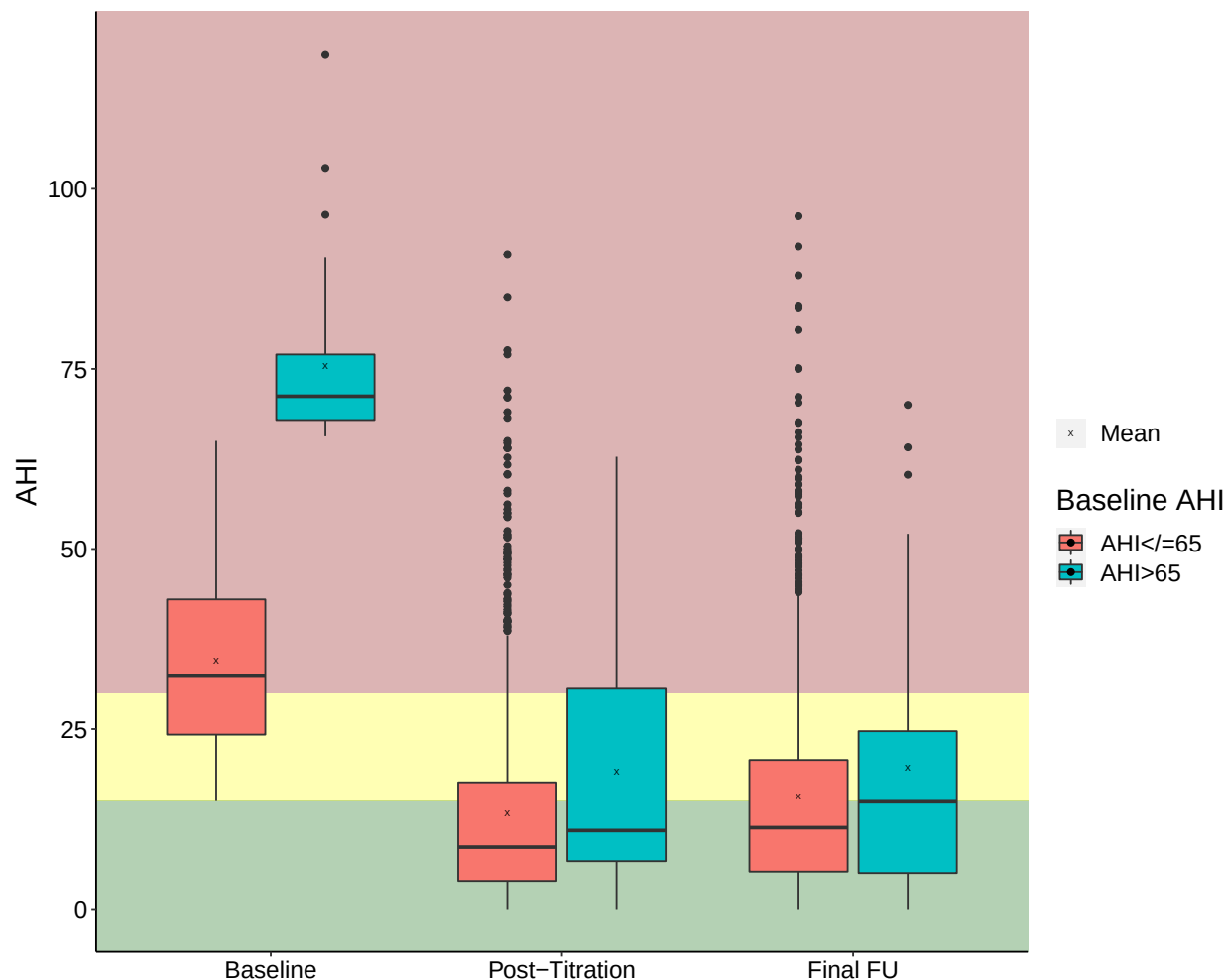
Table 2: AHI Outcomes by Baseline AHI

Variable	AHI $\leq$ 65	AHI $>$ 65	p-value	Type of Test
Final AHI	15.6 $\pm$ 14.8 (11.3), 0 - 96.2, N=1138	19.7 $\pm$ 19.46 (14.9), 0 - 70, N=41	0.4476	Wilcoxon Test
Change in AHI - Baseline to Final	18.86 $\pm$ 16.89 (18.5), -46.3 - 64.9, N=1138	55.67 $\pm$ 21.25 (60.8), 4.8 - 103.2, N=41	0.0000	Students t-test
Nonresponder - Sher	35.7% (406)	34.1% (14)	1.0000	Fisher's Exact Test
Responder - Sher	64.3% (732)	65.9% (27)	1.0000	

Note:

Format for numeric variables: Mean $\pm$ SD (Median), Range

There is no significant difference between AHI groups for Final AHI and for Response Rate. Change in AHI is significantly higher in patients with high AHI. 11 (26.83%) patients with baseline AHI $>$ 65 had no OSA (AHI $\leq$ 5.0) at the Final visit, 10 (24.39%) had mild OSA, 11 (26.83%) moderate, and 9 (21.95%) severe OSA at the Final Visit.



Epworth Sleepiness Scale (ESS)

Table 3: ESS Outcomes by Baseline AHI

Variable	AHI≤65	AHI>65	p-value	Type of Test
Final ESS	6.88±4.54 (6), 0 - 23, N=1231	6.54±4.86 (5.5), 0 - 20, N=46	0.4371	Wilcoxon Test
Change in ESS - Baseline to Final	4.53±5.21 (4), -13 - 23, N=1108	3.79±4.9 (4), -6 - 20, N=43	0.3407	Students t-test

Note:

Format for numeric variables: Mean±SD (Median), Range

There is no significant difference between the two AHI groups in regard to Final ESS and Change in ESS from Baseline to Final. However, as can be seen in the graph and table below, we are not able to say that the change in ESS for patients with AHI>65 is non-inferior to that of patients with AHI≤65 as the confidence interval for the difference between the two groups is above the margin of acceptability (ESS Change=2)

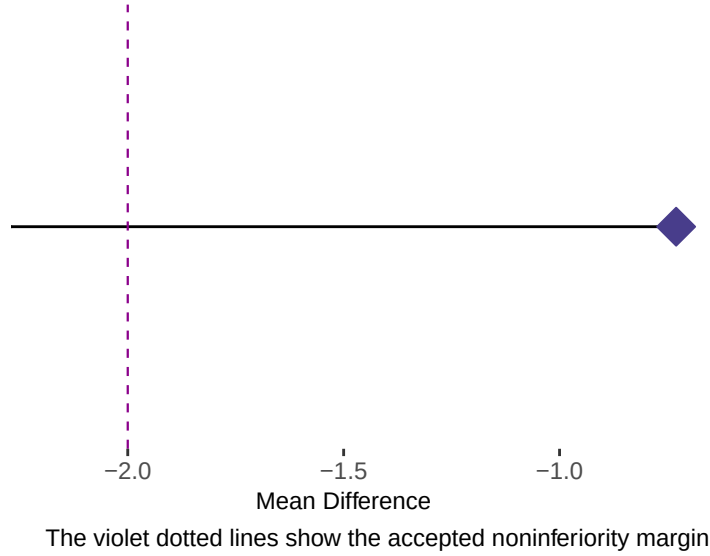


Figure 1: Visual of Non-Inferiority analysis for Change in ESS

Table 4: Non-inferiority Analysis for Change in ESS from Baseline to Final Follow-up

Group	Change in ESS (mean $\pm$ SD)	N	Mean Difference	Lower Bound of Confidence Interval	Accepted Lower Bound	p value
AHI $\leq$ 65	4.53 $\pm$ 5.21	1108	-0.73	-2.27	-2	0.05
AHI>65	3.79 $\pm$ 4.9	43				0.05

Usage

Table 5: Usage at Final Visit by Baseline AHI

Variable	AHI $\leq$ 65	AHI>65	p-value	Type of Test
Therapy Use at Final Visit	5.74 $\pm$ 2.21 (6), 0 - 10.29, N=1146	4.86 $\pm$ 2.57 (5.14), 0 - 9, N=43	0.0306	Students t-test

Note:

Format for numeric variables: Mean $\pm$ SD (Median), Range

There is a significant difference in the usage (hr/night) between the two AHI groups. Patients with AHI>65 use their device 52 minutes less per night on average than patients with AHI $\leq$ 65.

Clinical Global Impression - Improvement Scale

Table 6: Difference in CGI-I Classification between AHI Subgroups

CGI-I	AHI $\leq$ 65	AHI $>$ 65	p-value	Type of Test
1- Very much improved	N=446 (40.33%)	N=13 (28.89%)	0.197	Fisher's Exact Test (Improved vs Worse/No Change)
2 - Much improved	N=397 (35.9%)	N=15 (33.33%)		
3 - Minimally improved	N=157 (14.2%)	N=10 (22.22%)		
4 - No change	N=68 (6.15%)	N=6 (13.33%)		
5 - Minimally worse	N=24 (2.17%)	N=0 (0%)		
6 - Much worse	N=12 (1.08%)	N=1 (2.22%)		
7 - Very much worse	N=2 (0.18%)	N=0 (0%)		

Additional Analysis based on Deficiency Letter

Regressions based on Dr. Thaler's paper

Within the deficiency letter, it was suggested that in previous papers, AHI was a predictor of response when response was defined as AHI $<$ 10 or AHI $<$ 10 & 50% Reduction. The tables below show that this remains true with the data used for the submission

Table 7: Univariate Logistic Regression for Responder Rate (AHI $<$ 10) by Baseline AHI

	Estimate	Std. Error	OR (95% CI)	z value	p-value
(Intercept)	0.736	0.181	2.088 (1.47, 2.99)	4.071	4.7e-05
AHI.Pre.Enrollment	-0.026	0.005	0.975 (0.97, 0.98)	-5.374	7.7e-08

Table 8: Full Logistic Regression Model for Responder Rate (AHI $<$ 10) by Baseline AHI

	Estimate	Std. Error	OR (95% CI)	z value	p-value
(Intercept)	1.751	0.738	5.762 (1.36, 24.73)	2.372	1.8e-02
AHI.Pre.Enrollment	-0.024	0.005	0.977 (0.97, 0.99)	-4.924	8.5e-07
Age	0.000	0.006	1 (0.99, 1.01)	-0.017	9.9e-01
BMI.Pre.Enrollment	-0.033	0.018	0.967 (0.93, 1)	-1.817	6.9e-02
Final.FU.Therapy.Use	0.042	0.032	1.043 (0.98, 1.11)	1.317	1.9e-01
GenderMale	-0.452	0.165	0.636 (0.46, 0.88)	-2.742	6.1e-03

Table 9: Reduced Logistic Regression Model for Responder Rate (AHI $<$ 10) by Baseline AHI

	Estimate	Std. Error	OR (95% CI)	z value	p-value
(Intercept)	2.102	0.552	8.185 (2.8, 24.41)	3.810	1.4e-04
AHI.Pre.Enrollment	-0.024	0.005	0.976 (0.97, 0.99)	-4.990	6.0e-07
BMI.Pre.Enrollment	-0.037	0.018	0.964 (0.93, 1)	-2.027	4.3e-02
GenderMale	-0.464	0.162	0.629 (0.46, 0.86)	-2.858	4.3e-03

Table 10: Univariate Logistic Regression for Responder Rate (AHI<10 and 50 Percent AHI Reduction) by Baseline AHI

	Estimate	Std. Error	OR (95% CI)	z value	p-value
(Intercept)	0.660	0.180	1.934 (1.36, 2.76)	3.664	2.5e-04
AHI.Pre.Enrollment	-0.024	0.005	0.976 (0.97, 0.99)	-5.053	4.3e-07

Table 11: Full Logistic Regression Model for Responder Rate (AHI<10 and 50 Percent AHI Reduction) by Baseline AHI

	Estimate	Std. Error	OR (95% CI)	z value	p-value
(Intercept)	1.589	0.736	4.9 (1.16, 20.93)	2.159	3.1e-02
AHI.Pre.Enrollment	-0.022	0.005	0.978 (0.97, 0.99)	-4.605	4.1e-06
Age	0.001	0.006	1.001 (0.99, 1.01)	0.161	8.7e-01
BMI.Pre.Enrollment	-0.035	0.018	0.966 (0.93, 1)	-1.905	5.7e-02
Final.FU.Therapy.Use	0.044	0.032	1.045 (0.98, 1.11)	1.406	1.6e-01
GenderMale	-0.392	0.164	0.676 (0.49, 0.93)	-2.387	1.7e-02

Table 12: Reduced Logistic Regression Model for Responder Rate (AHI<10 and 50 Percent AHI Reduction) by Baseline AHI

	Estimate	Std. Error	OR (95% CI)	z value	p-value
(Intercept)	1.655	0.612	5.234 (1.59, 17.53)	2.704	6.9e-03
AHI.Pre.Enrollment	-0.022	0.005	0.978 (0.97, 0.99)	-4.602	4.2e-06
BMI.Pre.Enrollment	-0.035	0.018	0.966 (0.93, 1)	-1.919	5.5e-02
Final.FU.Therapy.Use	0.045	0.031	1.046 (0.98, 1.11)	1.432	1.5e-01
GenderMale	-0.396	0.162	0.673 (0.49, 0.92)	-2.441	1.5e-02

However, when looking at the different OSA severities, Baseline AHI remains significant only for patients with moderate OSA and only when the response is defined as AHI<10. See tables below. This could suggest that AHI is only a significant predictor when patients are in the moderate OSA group. Once baseline AHI is greater than 30 (or patients are in the severe group), the response rate is not correlated with baseline AHI.

Table 13: Univariate Logistic Regression for Responder Rate (AHI<10) by Baseline AHI for moderate OSA patients

	Estimate	Std. Error	OR (95% CI)	z value	p-value
(Intercept)	1.499	0.610	4.477 (1.37, 15.02)	2.457	1.4e-02
AHI.Pre.Enrollment	-0.055	0.026	0.947 (0.9, 1)	-2.088	3.7e-02

Table 14: Univariate Logistic Regression for Responder Rate (AHI<10) by Baseline AHI for severe OSA patients

	Estimate	Std. Error	OR (95% CI)	z value	p-value
(Intercept)	0.096	0.332	1.1 (0.58, 2.13)	0.288	7.7e-01
AHI.Pre.Enrollment	-0.013	0.007	0.987 (0.97, 1)	-1.758	7.9e-02

Table 15: Univariate Logistic Regression for Responder Rate (AHI<10 and 50 Percent AHI Reduction) by Baseline AHI for moderate OSA patients

	Estimate	Std. Error	OR (95% CI)	z value	p-value
(Intercept)	1.077	0.603	2.935 (0.91, 9.68)	1.784	7.4e-02
AHI.Pre.Enrollment	-0.038	0.026	0.962 (0.91, 1.01)	-1.469	1.4e-01

Table 16: Univariate Logistic Regression for Responder Rate (AHI<10 and 50 Percent AHI Reduction) by Baseline AHI for severe OSA patients

	Estimate	Std. Error	OR (95% CI)	z value	p-value
(Intercept)	0.096	0.332	1.1 (0.58, 2.13)	0.288	7.7e-01
AHI.Pre.Enrollment	-0.013	0.007	0.987 (0.97, 1)	-1.758	7.9e-02

The response rate for both definitions is significantly different between the severity groups, but is not significant between on and off label groups. Response rate (AHI<10) for moderate OSA patients is 56.14% and 38.41% for severe patients which is significant (chi-square test, p-value = 9.4e-08). When looking at the same response rate for on and off label patients (AHI<=65 vs AHI>65), the response rate was no longer significant (chi-square test, p-value = 3.7e-01) with a response rate of 45.8% for patients with AHI<=65 and 36.36% for patients with AHI>65.

Response rate (AHI<10 & 50% AHI Reduction) for moderate OSA patients is 55.09% and 38.41% for severe patients which is significant (chi-square test, p-value = 5.1e-07). When looking at the same response rate for on and off label patients (AHI<=65 vs AHI>65), the response rate was no longer significant (chi-square test, p-value = 4e-01) with a response rate of 45.37% for patients with AHI<=65 and 36.36% for patients with AHI>65.

STAR Response Rate for AHI>65

The response rate for patients with AHI>65 is 65.9% which is greater than 50%, as specified by STAR.

Improvements in Severity by AHI group (AHI<=65 vs AHI>65)

78% of patients with AHI>65 improved by at least one OSA severity category. This is similar to patients with AHI<=65 (78.3%)

Additional Analyses per FDA Meeting on 4/11/2022

Differences in US vs OUS

Table 17: Outcomes by Region

Variable	EU	US	p-value
N	588 (38.1%)	954 (61.9%)	
Age	56.85±10.02 (57), 21 - 81, N=588	62.19±10.43 (64), 28 - 85, N=947	0.0000
AHI.Pre.Enrollment	35.26±12.84 (32.8), 15.1 - 79.2, N=588	36.71±15.89 (33.9), 15 - 118.7, N=954	0.0500
BMI.Pre.Enrollment	28.85±3.66 (28.73), 19.88 - 42.78, N=577	29.18±3.82 (29.29), 17.71 - 47.19, N=931	0.0950
ESS.Total.Pre.Enrollment	11.93±5.33 (12), 0 - 24, N=561	10.9±5.66 (11), 0 - 24, N=791	0.0007

Based on the table above, there is no significant difference in baseline AHI or BMI between US centers and OUS centers. However, a trend shows that OUS sites have a lower baseline AHI than US sites. Additionally, patients at OUS sites are younger and have a significantly higher ESS at Baseline as compared to US centers.

Centers prescribing Inspire to off-label patients

Table 18: Sites and their contribution to on/off - label patients for AHI

Site	Percent of AHI</=65 patients in ADHERE that were enrolled at this Site	Percent of AHI>65 patients in ADHERE that were enrolled at this Site	Percent of Patients at this site with AHI>65
Cincinnati VA	35 (2.4%)	9 (15.3%)	9 (20.5%)
Thomas Jefferson University and Hospital	63 (4.2%)	6 (10.2%)	6 (8.7%)
University of Pennsylvania	22 (1.5%)	5 (8.5%)	5 (18.5%)
Technischen Universität München	135 (9.1%)	4 (6.8%)	4 (2.9%)
Medical College of Wisconsin	31 (2.1%)	3 (5.1%)	3 (8.8%)
The Ohio State University	18 (1.2%)	3 (5.1%)	3 (14.3%)
Universitäts-HNO-Klinik Mannheim	85 (5.7%)	3 (5.1%)	3 (3.4%)
University of Alabama at Birmingham	35 (2.4%)	3 (5.1%)	3 (7.9%)
Kansas University Medical Center	57 (3.8%)	2 (3.4%)	2 (3.4%)
Kantonsspital Baselland	35 (2.4%)	2 (3.4%)	2 (5.4%)
OLVG	59 (4%)	2 (3.4%)	2 (3.3%)
Richard L Roudebush VA Medical Center	12 (0.8%)	2 (3.4%)	2 (14.3%)
University Hospital Cleveland	13 (0.9%)	2 (3.4%)	2 (13.3%)
University of Pittsburgh Medical Center	99 (6.7%)	2 (3.4%)	2 (2%)
Cleveland Clinic Foundation	58 (3.9%)	1 (1.7%)	1 (1.7%)
Medical University of South Carolina	6 (0.4%)	1 (1.7%)	1 (14.3%)
MetroHealth	12 (0.8%)	1 (1.7%)	1 (7.7%)
St. Cloud Ear, Nose & Throat	30 (2%)	1 (1.7%)	1 (3.2%)
St. Luke Presbyterian Hospital	13 (0.9%)	1 (1.7%)	1 (7.1%)
Technische Universität Dresden	5 (0.3%)	1 (1.7%)	1 (16.7%)
The University of Iowa	9 (0.6%)	1 (1.7%)	1 (10%)
The University of Texas Southwestern Medical Center	43 (2.9%)	1 (1.7%)	1 (2.3%)

University of Tennessee Health Science Center	5 (0.3%)	1 (1.7%)	1 (16.7%)
Vanderbilt University Medical Center	23 (1.6%)	1 (1.7%)	1 (4.2%)
Weill Cornell Medicine	56 (3.8%)	1 (1.7%)	1 (1.8%)
Advanced ENT and Allergy	28 (1.9%)	0 (0%)	0 (0%)
Asklepios Kliniken Hamburg	16 (1.1%)	0 (0%)	0 (0%)
Hackensack University Medical Center	25 (1.7%)	0 (0%)	0 (0%)
Indiana University	28 (1.9%)	0 (0%)	0 (0%)
Klinik für HNO-Heilkunde/HNO-Schlaflabor	170 (11.5%)	0 (0%)	0 (0%)
Klinikum Stuttgart-Katharinenhospital	8 (0.5%)	0 (0%)	0 (0%)
Mayo Clinic Arizona	1 (0.1%)	0 (0%)	0 (0%)
MedStar Health Research Institute	1 (0.1%)	0 (0%)	0 (0%)
Middlesex Hospital	7 (0.5%)	0 (0%)	0 (0%)
Northshore University	15 (1%)	0 (0%)	0 (0%)
Northwell Health	4 (0.3%)	0 (0%)	0 (0%)
Ogden Clinic	20 (1.3%)	0 (0%)	0 (0%)
Pennsylvania Hospital	14 (0.9%)	0 (0%)	0 (0%)
Puget Sound ENT	23 (1.6%)	0 (0%)	0 (0%)
Reading Hospital	2 (0.1%)	0 (0%)	0 (0%)
Rush University Medical Center	9 (0.6%)	0 (0%)	0 (0%)
Sint Antonius Hospital	50 (3.4%)	0 (0%)	0 (0%)
Sleep Management Institute	5 (0.3%)	0 (0%)	0 (0%)
St. Joseph Mercy Hospital	4 (0.3%)	0 (0%)	0 (0%)
Stanford University Medical Center	2 (0.1%)	0 (0%)	0 (0%)
Universitair Ziekenhuis Antwerpen	7 (0.5%)	0 (0%)	0 (0%)
Universitätsklinikum Hamburg Eppendorf	6 (0.4%)	0 (0%)	0 (0%)
University of Colorado Denver	24 (1.6%)	0 (0%)	0 (0%)
University of Minnesota Fairview	11 (0.7%)	0 (0%)	0 (0%)
University of South Florida	9 (0.6%)	0 (0%)	0 (0%)
University of Southern California	10 (0.7%)	0 (0%)	0 (0%)
Valley ENT	19 (1.3%)	0 (0%)	0 (0%)
Valley Sleep Center	1 (0.1%)	0 (0%)	0 (0%)
Virginia Commonwealth University	5 (0.3%)	0 (0%)	0 (0%)
Total N	1483	59	

25 (46.3%) sites out of 54 have implanted and provided follow-up data on at least one AHI off-label patient. No site had a majority of these patients, however, Cincinnati VA and Thomas Jefferson University and Hospital account for 25.4% of the patients with AHI>65.

Table 19: Baseline and Outcome Variables for High Enrolling Sites of AHI>65 patients vs Other Off-Label Enrolling Sites

Variable	High Enroller (Cincinnati and TJU)	Non-High Enroller (All other sites with any AHI>65)	p-value
Baseline AHI	78.63±9.98 (80), 65.64-102.9, N=15	74.83±9.9 (71.9), 65.8-118.7, N=44	0.2138
Baseline ESS	11.07±5.36 (11), 2-21, N=14	10.53±5.54 (10), 0-22, N=40	0.7476
Baseline BMI	30.94±3.79 (31.94), 22.43-35.44, N=14	30.55±3.51 (30.62), 21.8-38.35, N=44	0.7327
Change in AHI (BL - Final)	55.51±10.38 (52.7), 42.6-69.9, N=7	55.71±22.97 (61.45), 4.8-103.2, N=34	0.9712
Final AHI*	21.53±19.16 (17.3), 3.1-60.3, N=7	19.33±19.79 (11.8), 0-70, N=34	0.5443

Note:

Format for numeric variables: Mean±SD (Median) Range

p-value derived from Student's t-test except where otherwise specified

*p-value derived from Wilcoxon Rank-Sum Test

Table 20: Sites and their contribution to on/off - label patients for BMI

Site	Percent of BMI</=32 patients in ADHERE that were enrolled at this Site	Percent of BMI>32 patients in ADHERE that were enrolled at this Site	Percent of Patients at this site with BMI>32
Technischen Universität München	100 (8.2%)	39 (13.4%)	39 (28.1%)
Klinik für HNO-Heilkunde/HNO-Schlaf Labor	136 (11.2%)	33 (11.4%)	33 (19.5%)
Universitäts-HNO-Klinik Mannheim	58 (4.8%)	28 (9.7%)	28 (32.6%)
Cincinnati VA	25 (2.1%)	16 (5.5%)	16 (39%)
Thomas Jefferson University and Hospital	51 (4.2%)	14 (4.8%)	14 (21.5%)
University of Alabama at Birmingham	25 (2.1%)	13 (4.5%)	13 (34.2%)
Weill Cornell Medicine	44 (3.6%)	13 (4.5%)	13 (22.8%)
The University of Texas Southwestern Medical Center	34 (2.8%)	10 (3.4%)	10 (22.7%)
University of Pittsburgh Medical Center	91 (7.5%)	10 (3.4%)	10 (9.9%)
Indiana University	18 (1.5%)	9 (3.1%)	9 (33.3%)
The Ohio State University	12 (1%)	9 (3.1%)	9 (42.9%)
Cleveland Clinic Foundation	51 (4.2%)	8 (2.8%)	8 (13.6%)
Medical College of Wisconsin	23 (1.9%)	8 (2.8%)	8 (25.8%)
University of Pennsylvania	20 (1.6%)	7 (2.4%)	7 (25.9%)
Kansas University Medical Center	54 (4.4%)	5 (1.7%)	5 (8.5%)
Advanced ENT and Allergy	24 (2%)	4 (1.4%)	4 (14.3%)
Hackensack University Medical Center	21 (1.7%)	4 (1.4%)	4 (16%)
MetroHealth	9 (0.7%)	4 (1.4%)	4 (30.8%)
Valley ENT	15 (1.2%)	4 (1.4%)	4 (21.1%)

Vanderbilt University Medical Center	20 (1.6%)	4 (1.4%)	4 (16.7%)
Kantonsspital Baselland	33 (2.7%)	3 (1%)	3 (8.3%)
Ogden Clinic	16 (1.3%)	3 (1%)	3 (15.8%)
OLVG	58 (4.8%)	3 (1%)	3 (4.9%)
Pennsylvania Hospital	5 (0.4%)	3 (1%)	3 (37.5%)
Richard L Roudebush VA Medical Center	11 (0.9%)	3 (1%)	3 (21.4%)
St. Cloud Ear, Nose & Throat	28 (2.3%)	3 (1%)	3 (9.7%)
St. Luke Presbyterian Hospital	11 (0.9%)	3 (1%)	3 (21.4%)
University of Colorado Denver	21 (1.7%)	3 (1%)	3 (12.5%)
University of South Florida	5 (0.4%)	3 (1%)	3 (37.5%)
Northshore University	13 (1.1%)	2 (0.7%)	2 (13.3%)
Rush University Medical Center	7 (0.6%)	2 (0.7%)	2 (22.2%)
Technische Universität Dresden	4 (0.3%)	2 (0.7%)	2 (33.3%)
University Hospital Cleveland	12 (1%)	2 (0.7%)	2 (14.3%)
University of Southern California	8 (0.7%)	2 (0.7%)	2 (20%)
Virginia Commonwealth University	3 (0.2%)	2 (0.7%)	2 (40%)
Asklepios Kliniken Hamburg	15 (1.2%)	1 (0.3%)	1 (6.2%)
Middlesex Hospital	6 (0.5%)	1 (0.3%)	1 (14.3%)
Puget Sound ENT	20 (1.6%)	1 (0.3%)	1 (4.8%)
Reading Hospital	1 (0.1%)	1 (0.3%)	1 (50%)
Sleep Management Institute	4 (0.3%)	1 (0.3%)	1 (20%)
St. Joseph Mercy Hospital	3 (0.2%)	1 (0.3%)	1 (25%)
The University of Iowa	9 (0.7%)	1 (0.3%)	1 (10%)
Universitätsklinikum Hamburg Eppendorf	5 (0.4%)	1 (0.3%)	1 (16.7%)
University of Tennessee Health Science Center	5 (0.4%)	1 (0.3%)	1 (16.7%)
Klinikum Stuttgart-Katharinenhospital	8 (0.7%)	0 (0%)	0 (0%)
Mayo Clinic Arizona	1 (0.1%)	0 (0%)	0 (0%)
Medical University of South Carolina	7 (0.6%)	0 (0%)	0 (0%)
MedStar Health Research Institute	1 (0.1%)	0 (0%)	0 (0%)
Northwell Health	3 (0.2%)	0 (0%)	0 (0%)
Sint Antonius Hospital	44 (3.6%)	0 (0%)	0 (0%)
Stanford University Medical Center	2 (0.2%)	0 (0%)	0 (0%)
Universitair Ziekenhuis Antwerpen	6 (0.5%)	0 (0%)	0 (0%)
University of Minnesota Fairview	11 (0.9%)	0 (0%)	0 (0%)
Valley Sleep Center	1 (0.1%)	0 (0%)	0 (0%)
Total N	1218	290	

44 (81.48%) sites out of 54 have implanted and provided follow-up data on at least one BMI off-label patient. No site had a majority of these patients, however, the Technical University of Munich and Lubeck account for 24.8% of the patients with BMI>32.

Table 21: Baseline and Outcome Variables for High Enrolling Sites of BMI>32 patients vs Other Off-Label Enrolling Sites

Variable	High Enroller (TUM and Lubeck)	Non-High Enroller (All other sites with any BMI>32)	p-value
Baseline AHI	36.31±14.3 (33.9), 15.1-70.6, N=72	39.96±17.59 (37.85), 15.6-118.7, N=218	0.0790
Baseline ESS	12.54±5.08 (12), 1-24, N=67	10.72±5.7 (11), 0-24, N=190	0.0162
Baseline BMI	34.42±1.99 (33.95), 32.19-42.78, N=72	34.33±2.58 (33.59), 32.01-47.19, N=218	0.7761
Change in AHI (BL - Final)	20.94±16.3 (21.55), -27.5-61.2, N=70	20.37±20.7 (18.3), -46.3-103.2, N=154	0.8239
Final AHI*	15.38±14.23 (12.2), 0.3-92, N=70	18.31±17.86 (13.35), 0-96.2, N=154	0.4595

Note:

Format for numeric variables: Mean±SD (Median) Range

p-value derived from Student's t-test except where otherwise specified

*p-value derived from Wilcoxon Rank-Sum Test

There is a significant difference in baseline ESS between the high enrolling sites for high BMI patients compared to the rest of sites who have enrolled any high BMI patients. However, this difference may be due to the difference in the regions. In the sections above, we saw that patients at centers in Europe have a higher baseline ESS than patients at centers in the US.

Medicare Patients

Table 22: Medicare patients by label

Label	Medicare	No Medicare	p-value
AHI≤65	295 (19.9%)	1188 (80.1%)	0.87
AHI>65	12 (20.3%)	47 (79.7%)	
BMI≤32	233 (19.1%)	985 (80.9%)	0.09
BMI>32	69 (23.8%)	221 (76.2%)	

There is no significant difference in Medicare patients between AHI on/off label patients and BMI on/off label patients.

Oral Appliance

Table 23: Demographic Variables by Previous Treatment

Variable	CPAP only	CPAP and Oral Appliance	No CPAP or Oral Appliance	Oral Appliance only	p-value
N	1161 (75.7%)	340 (22.2%)	23 (1.5%)	9 (0.6%)	
Age	60.14±10.62 (61), 28-85, N=1156	60.41±10.34 (61), 30-85, N=340	57.55±14.27 (59), 21-80, N=22	59±9.27 (59), 42-73, N=9	0.6532
AHI - Baseline	36.18±15.09 (33), 15-118.7, N=1161	36.1±14.07 (33.95), 15.1-96.4, N=340	34.24±12.49 (33), 17-60.5, N=23	36.28±13.3 (39.1), 19-54.19, N=9	0.9421
BMI - Baseline	29.23±3.7 (29.25), 17.71-47.19, N=1139	28.59±3.84 (28.56), 18.38-45.73, N=330	28.5±4.31 (28.77), 21.29-35.15, N=22	27.23±5.53 (29.29), 18.88-34.79, N=9	0.0187
ESS - Baseline	11.49±5.49 (12), 0-24, N=1008	11.05±5.7 (11), 0-24, N=310	9.29±5.73 (9), 0-17, N=17	8.75±6.48 (6), 3-20, N=8	0.1272

Note:

Format for numeric variables: Mean±SD (Median) Range

1161 (75.7%) of patients have only attempted CPAP prior to Inspire. Only 349 (23%) of patients have received Oral Appliance either with or without CPAP.

There are no differences in OSA severity between the patient groups.

Additional Analyses based on Deficiency Letter - January 2023

Outcomes by Region and Site

Table 24: Outcomes for patients with 65<AHI<=100 By Region

Variable	OUS	US	p-value
Final AHI*	26.6±19.3 (24.15), 6.5 - 64.1, N=11	15.7±18.04 (10.6), 0 - 70, N=28	0.0379
Change in AHI - Baseline to Final	43.91±21.33 (43.35), 4.8 - 72.7, N=11	59.07±18.49 (63), 17 - 92.3, N=28	0.0547
Final Sher Response Rate	45.5% (5)	75% (21)	0.1306
Final ESS*	7.5±6.55 (6), 1 - 20, N=10	6.03±4.15 (5), 0 - 16, N=35	0.8263
Change in ESS - Baseline to Final	0.1±3.14 (1), -5 - 4, N=10	5.19±4.61 (4.5), -6 - 20, N=32	0.0007
Therapy Use (hr/night) at Final Visit	5.43±1.83 (5.36), 2.86 - 8.71, N=10	4.84±2.76 (5.14), 0 - 9, N=31	0.4472

Note:

p-value derived using Student's t-test and Fisher's Exact test except where otherwise specified

*p-value derived using Wilcoxon Rank Sum Test due to non-normal distribution

Table 25: Outcomes for patients with $65 < \text{AHI} \leq 100$ By Site

Site Name	N	Final AHI	Change in AHI - Baseline to Final	Final Sher Response	Final ESS	Change in ESS - Baseline to Final	Therapy Use (hr/night) at Final Visit
Cincinnati VA	8	21.87±5.73 (20), 17.3 - 28.3, N=3	53.37±6.63 (52.7), 47.1 - 60.3, N=3	66.7% (2)	5.88±3.64 (6.5), 1 - 11, N=8	4.88±7.61 (3.5), -6 - 20, N=8	6.52±2.14 (6.86), 4.14 - 9, N=6
Thomas Jefferson University and Hospital	6	8.27±7.29 (5.1), 3.1 - 16.6, N=3	61.95±11.28 (66.9), 49.04 - 69.9, N=3	100% (3)	11.33±4.04 (9), 9 - 16, N=3	2.67±2.08 (2), 1 - 5, N=3	6.86±1.03 (6.57), 6 - 8, N=3
University of Pennsylvania	5	17.03±23.67 (7.2), 1.6 - 52.1, N=4	60.3±31.64 (65.3), 18.3 - 92.3, N=4	75% (3)	4.75±1.5 (5), 3 - 6, N=4	4.33±1.53 (4), 3 - 6, N=3	2.46±2.16 (2.07), 0.57 - 5.14, N=4
Technischen Universität München	4	24.9±17.44 (22.2), 9.4 - 45.8, N=4	45.5±19.81 (47.35), 23.9 - 63.4, N=4	50% (2)	9.25±7.63 (7.5), 2 - 20, N=4	-1±3.37 (-2.5), -3 - 4, N=4	4.46±1.12 (4.57), 3 - 5.71, N=4
Medical College of Wisconsin	3	32.95±25.53 (32.95), 14.9 - 51, N=2	36.55±23.41 (36.55), 20 - 53.1, N=2	50% (1)	8.33±4.51 (8), 4 - 13, N=3	3.33±2.08 (4), 1 - 5, N=3	3.43±NA (3.43), 3.43 - 3.43, N=1
The Ohio State University	3	11.7±8.23 (10.9), 3.9 - 20.3, N=3	59.97±7.78 (60.8), 51.8 - 67.3, N=3	66.7% (2)	4±3 (4), 1 - 7, N=3	6±2.65 (7), 3 - 8, N=3	3.57±2.06 (4.14), 1.29 - 5.29, N=3
Universitäts-HNO-Klinik Mannheim	3	44.4±27.86 (44.4), 24.7 - 64.1, N=2	23.6±26.59 (23.6), 4.8 - 42.4, N=2	0% (0)	8±NA (8), 8 - 8, N=1	-5±NA (-5), -5 - -5, N=1	2.86±NA (2.86), 2.86 - 2.86, N=1
University of Alabama at Birmingham	3	41.7±NA (41.7), 41.7 - 41.7, N=1	35.1±NA (35.1), 35.1 - 35.1, N=1	0% (0)	6.33±2.89 (8), 3 - 8, N=3	6.67±2.52 (7), 4 - 9, N=3	4.43±5.05 (4.43), 0.86 - 8, N=2
Kansas University Medical Center	2	5.5±0.71 (5.5), 5 - 6, N=2	66.55±6.86 (66.55), 61.7 - 71.4, N=2	100% (2)	N=0	N=0	5.14±NA (5.14), 5.14 - 5.14, N=1
Kantonsspital Baselland	2	21±4.45 (21), 17.85 - 24.15, N=2	46.7±4.74 (46.7), 43.35 - 50.05, N=2	50% (1)	10.5±10.61 (10.5), 3 - 18, N=2	3±1.41 (3), 2 - 4, N=2	7±0 (7), 7 - 7, N=2
OLVG	2	27.85±30.05 (27.85), 6.6 - 49.1, N=2	43.85±31.04 (43.85), 21.9 - 65.8, N=2	50% (1)	2±1.41 (2), 1 - 3, N=2	1.5±0.71 (1.5), 1 - 2, N=2	5.43±0.61 (5.43), 5 - 5.86, N=2

Table 25: Outcomes for patients with $65 < \text{AHI} \leq 100$ By Site
(continued)

Site Name	N	Final AHI	Change in AHI - Baseline to Final	Final Sher Response	Final ESS	Change in ESS - Baseline to Final	Therapy Use (hr/night) at Final Visit
Richard L Roudebush VA Medical Center	2	N=0	N=0	N=0	2±2.83 (2), 0 - 4, N=2	3.5±4.95 (3.5), 0 - 7, N=2	0.36±0.3 (0.36), 0.14 - 0.57, N=2
University Hospital Cleveland	2	0±0 (0), 0 - 0, N=2	76.8±0 (76.8), 76.8 - 76.8, N=2	100% (2)	15±NA (15), 15 - 15, N=1	4±NA (4), 4 - 4, N=1	0±NA (0), 0 - 0, N=1
University of Pittsburgh Medical Center	2	1.8±0.71 (1.8), 1.3 - 2.3, N=2	69.7±7.07 (69.7), 64.7 - 74.7, N=2	100% (2)	7.5±6.36 (7.5), 3 - 12, N=2	9±7.07 (9), 4 - 14, N=2	6.36±1.92 (6.36), 5 - 7.71, N=2
Cleveland Clinic Foundation	1	0.9±NA (0.9), 0.9 - 0.9, N=1	78.4±NA (78.4), 78.4 - 78.4, N=1	100% (1)	N=0	N=0	5.43±NA (5.43), 5.43 - 5.43, N=1
Medical University of South Carolina	1	N=0	N=0	N=0	2±NA (2), 2 - 2, N=1	6±NA (6), 6 - 6, N=1	NaN±NA (NA), Inf - -Inf, N=0
MetroHealth	1	N=0	N=0	N=0	N=0	N=0	NaN±NA (NA), Inf - -Inf, N=0
St. Cloud Ear, Nose & Throat	1	17.4±NA (17.4), 17.4 - 17.4, N=1	73.1±NA (73.1), 73.1 - 73.1, N=1	100% (1)	N=0	N=0	5.14±NA (5.14), 5.14 - 5.14, N=1
St. Luke Presbyterian Hospital	1	11.9±NA (11.9), 11.9 - 11.9, N=1	54.1±NA (54.1), 54.1 - 54.1, N=1	100% (1)	4±NA (4), 4 - 4, N=1	N=0	7.14±NA (7.14), 7.14 - 7.14, N=1
Technische Universitat Dresden	1	6.5±NA (6.5), 6.5 - 6.5, N=1	72.7±NA (72.7), 72.7 - 72.7, N=1	100% (1)	5±NA (5), 5 - 5, N=1	1±NA (1), 1 - 1, N=1	8.71±NA (8.71), 8.71 - 8.71, N=1
The University of Iowa	1	0.8±NA (0.8), 0.8 - 0.8, N=1	65.6±NA (65.6), 65.6 - 65.6, N=1	100% (1)	1±NA (1), 1 - 1, N=1	N=0	NaN±NA (NA), Inf - -Inf, N=0
The University of Texas Southwestern Medical Center	1	22.7±NA (22.7), 22.7 - 22.7, N=1	64.3±NA (64.3), 64.3 - 64.3, N=1	0% (0)	10±NA (10), 10 - 10, N=1	3±NA (3), 3 - 3, N=1	7.86±NA (7.86), 7.86 - 7.86, N=1
University of Tennessee Health Science Center	1	N=0	N=0	N=0	2±NA (2), 2 - 2, N=1	9±NA (9), 9 - 9, N=1	8±NA (8), 8 - 8, N=1
Weill Cornell Medicine	1	70±NA (70), 70 - 70, N=1	17±NA (17), 17 - 17, N=1	0% (0)	2±NA (2), 2 - 2, N=1	11±NA (11), 11 - 11, N=1	5.29±NA (5.29), 5.29 - 5.29, N=1

Note:

N=0 indicates that no patients at this site had data entered for this variable

Comparison of Baseline Characteristics of Patients with vs without Follow-up

Patients were marked as having missed their final follow-up if their implant occurred more than 2 years prior to data download (7/25/2022). 3699 patients were implanted by 7/25/2022. 2399 of these patients were expected to have a final visit. 1542 (64.3%) completed the final visit.

Note that 5 out the 3699 patients had an unknown implant date. 3 of these patients completed a follow-up visit. The other 2 patients did not complete a final follow-up visit and were therefore left out of the analysis in the table below as it could not be determined when the patient's final follow-up should have occurred.

Table 26: Differences in Baseline Characteristics for patients who followed up vs those who did not

Variable	Completed Visit	Missed Visit	p-value
N	64.3% (1542)	35.7% (857)	
Age	60.14±10.59 (61), 21 - 85, N=1535	60.7±11.04 (62), 21 - 87, N=853	0.2341
Baseline AHI	36.16±14.82 (33.3), 15 - 118.7, N=1542	35.83±15.42 (32.8), 15 - 105.3, N=857	0.6183
Baseline BMI	29.06±3.76 (29.05), 17.71 - 47.19, N=1508	29.55±3.84 (29.58), 16.95 - 44.92, N=829	0.0026
Baseline ESS	11.33±5.55 (11), 0 - 24, N=1352	11.17±5.59 (11), 0 - 24, N=732	0.5254
Female	23.8% (366)	32.6% (278)	<0.001
White	95.7% (1462)	93.9% (785)	0.06
Hispanic or Latino	1.9% (29)	2.3% (19)	0.541
Prior Oral App Use	22.8% (349)	23.9% (204)	0.544
Prior CPAP Use	97.9% (1501)	96.5% (823)	0.043
AHI≤65	96.2% (1483)	96.6% (828)	
65<AHI≤75	2% (31)	1.6% (14)	
75<AHI≤100	1.7% (26)	1.6% (14)	0.933
AHI>100	0.1% (2)	0.1% (1)	

Note:

p-values derived using Student's t-test for numeric variables and Fisher's Exact Test for categorical variables

There was no significant difference in follow-up rates by Baseline AHI. Patients with a baseline AHI≤65 events/hr had a follow-up rate of 64.2%, patients with an AHI between 65 and ≤75 events/hr had a follow-up rate of 68.9%, patients with an AHI between 75 and ≤100 events/hr had a follow-up rate of 65% and patients with an AHI>100 events/hr had a follow-up rate of 66.7%



Label Expansion - BMI > 32

ADHERE Registry

Lauren Makey

02/03/23

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Demographics and Baseline Information

1508 patients enrolled into the ADHERE registry by July 22, 2022 had BMI collected at baseline (1218 with BMI $\leq$ 32 and 290 with BMI $>$ 32) and had a final follow up visit with at least one of the following values recorded at the final visit: AHI, ESS, Therapy Usage, CGI, and/or Patient Satisfaction data. Patients with higher BMI have a significantly higher AHI at Baseline (p-value = 0.001).

Table 1: Demographic Variables by Baseline BMI (BMI $\leq$ 32 vs. BMI $>$ 32)

Variable	BMI $\leq$ 32	BMI $>$ 32	p-value*
N	1218 (80.8%)	290 (19.2%)	
Age	60.34 $\pm$ 10.73 (61), 21 - 85, N=1212	59.42 $\pm$ 10.09 (60), 28 - 81, N=290	0.1697
AHI - Baseline	35.5 $\pm$ 14.26 (32.7), 15 - 102.9, N=1218	39.05 $\pm$ 16.89 (36.65), 15.1 - 118.7, N=290	0.0010
BMI - Baseline	27.79 $\pm$ 2.8 (28.18), 17.71 - 32, N=1218	34.35 $\pm$ 2.45 (33.67), 32.01 - 47.19, N=290	0.0000
ESS - Baseline	11.39 $\pm$ 5.52 (11), 0 - 24, N=1069	11.19 $\pm$ 5.59 (11), 0 - 24, N=257	0.6176
Female	24.1% (292)	23.6% (68)	0.9390
White	95.6% (1157)	96.1% (274)	0.8710
Hispanic or Latino	2.2% (26)	1.1% (3)	0.3390
Prior Oral App Use	23.6% (285)	18.6% (54)	0.0730
Prior CPAP Use	98.1% (1187)	97.2% (282)	0.3590

Note:

Format for numeric variables: Mean $\pm$ SD (Median), Range.

*P-value obtained using a Student's t-test for numeric variables and Fisher's exact test for categorical variables.

Table 2: Demographic Variables by Baseline BMI (BMI $\leq$ 32 vs. 32<BMI $\leq$ 35 vs. BMI>35)

Variable	BMI $\leq$ 32	32<BMI $\leq$ 35	BMI>35	p-value*
N	1218 (80.8%)	222 (14.7%)	68 (4.5%)	
Age	60.34 $\pm$ 10.73 (61), 21 - 85, N=1212	59.94 $\pm$ 10.04 (60), 28 - 81, N=222	57.75 $\pm$ 10.13 (58.5), 33 - 77, N=68	0.1378
AHI - Baseline	35.5 $\pm$ 14.26 (32.7), 15 - 102.9, N=1218	38.89 $\pm$ 16.54 (36.5), 15.5 - 118.7, N=222	39.58 $\pm$ 18.08 (37.45), 15.1 - 86, N=68	0.0012
BMI - Baseline	27.79 $\pm$ 2.8 (28.18), 17.71 - 32, N=1218	33.33 $\pm$ 0.86 (33.14), 32.01 - 34.97, N=222	37.69 $\pm$ 2.93 (36.6), 35.15 - 47.19, N=68	0.0000
ESS - Baseline	11.39 $\pm$ 5.52 (11), 0 - 24, N=1069	11.29 $\pm$ 5.66 (11.5), 0 - 24, N=196	10.89 $\pm$ 5.39 (11), 0 - 22, N=61	0.7777
Female	24.1% (292)	23.5% (52)	23.9% (16)	0.9920
White	95.6% (1157)	96.8% (210)	94.1% (64)	0.5670
Hispanic or Latino	2.2% (26)	0.9% (2)	1.5% (1)	0.4610
Prior Oral App Use	23.6% (285)	18% (40)	20.6% (14)	0.1850
Prior CPAP Use	98.1% (1187)	96.8% (215)	98.5% (67)	0.3840

Note:

Format for numeric variables: Mean $\pm$ SD (Median), Range.

*P-value comparing all three BMI groups, obtained using ANOVA for numeric variables and Fisher's exact test for categorical variables.

Outcomes at Final Visit

AHI

Table 3: AHI Outcomes by Baseline BMI (BMI $\leq$ 32 vs. BMI $>$ 32)

Variable	BMI $\leq$ 32	BMI $>$ 32	p-value	Type of Test
Final AHI	15.34 $\pm$ 14.44 (10.95), 0 - 83.8, N=930	17.4 $\pm$ 16.83 (12.95), 0 - 96.2, N=224	0.0714	Wilcoxon Test
Change in AHI - Baseline to Final	20.09 $\pm$ 18.11 (19.25), -41.6 - 92.3, N=930	20.55 $\pm$ 19.39 (19.35), -46.3 - 103.2, N=224	0.7478	Students t-test
Nonresponder - Sher	34.9% (325)	39.7% (89)	0.1880	Fisher's Exact Test
Responder - Sher	65.1% (605)	60.3% (135)		

Note:

Format for numeric variables: Mean $\pm$ SD (Median), Range

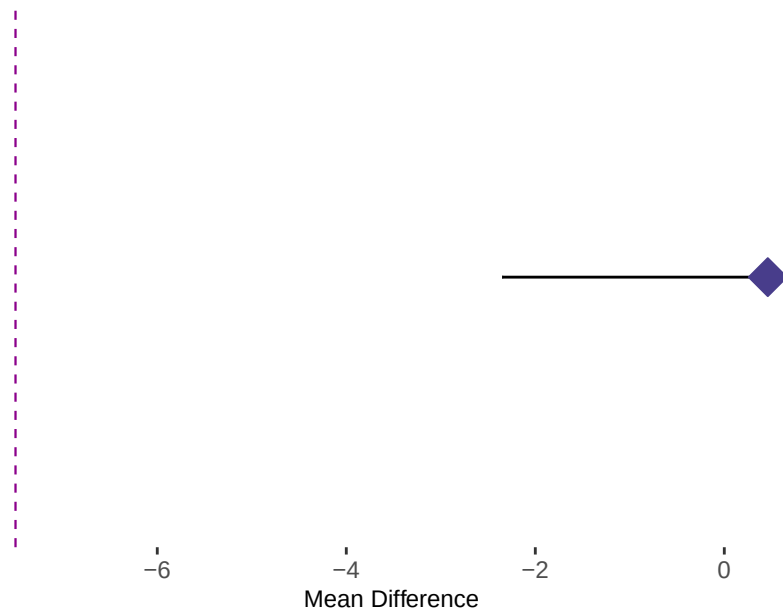
Table 4: AHI Outcomes by Baseline BMI (BMI $\leq$ 32 vs. 32<BMI $\leq$ 35 vs. BMI $>$ 35)

Variable	BMI $\leq$ 32	32<BMI $\leq$ 35	BMI $>$ 35	p-value	Type of Test
Final AHI	15.34 $\pm$ 14.44 (10.95), 0 - 83.8, N=930	17.39 $\pm$ 16.9 (12.95), 0 - 96.2, N=170	17.42 $\pm$ 16.79 (13.35), 0 - 92, N=54	0.1938	Kruskal-Wallis Test
Change in AHI - Baseline to Final	20.09 $\pm$ 18.11 (19.25), -41.6 - 92.3, N=930	21.15 $\pm$ 20.13 (19.95), -46.3 - 103.2, N=170	18.68 $\pm$ 16.91 (18.7), -27.5 - 56.2, N=54	0.6541	ANOVA
Nonresponder - Sher	34.9% (325)	39.4% (67)	40.7% (22)	0.3920	Fisher's Exact Test
Responder - Sher	65.1% (605)	60.6% (103)	59.3% (32)		

Note:

Format for numeric variables: Mean $\pm$ SD (Median), Range

There is no statistically significant difference between the BMI groups (BMI $\leq$ 32 vs. BMI $>$ 32 and BMI $\leq$ 32 vs. 32<BMI $\leq$ 35 vs. BMI $>$ 35) in regard to Final AHI, Change in AHI from Baseline to Final Visit, and Response Rate (at the 0.05 significance level). Additionally, as can be seen in the table and graph below, the change in AHI for patients with BMI $>$ 32 is non-inferior to that of patients with BMI $\leq$ 32, as the confidence interval for the difference between the two groups is within the margin of acceptability (AHI change = 7.5).



The violet dotted lines show the accepted noninferiority margin

Figure 1: Visual of Non-Inferiority analysis for Change in AHI (BMI $\leq$ 32 vs. BMI $>$ 32)

Table 5: Non-inferiority Analysis for Change in AHI from Baseline to Final Follow-up (BMI $\leq$ 32 vs. BMI $>$ 32)

Group	Change in AHI (mean $\pm$ SD)	N	Mean Difference	Lower Bound of Confidence Interval	Accepted Lower Bound	p-value
BMI $\leq$ 32	20.09 $\pm$ 18.11	930	0.46	-2.35	-7.5	0
BMI $>$ 32	20.55 $\pm$ 19.39	224				

Epworth Sleepiness Scale (ESS)

Table 6: ESS Outcomes by Baseline BMI (BMI $\leq$ 32 vs. BMI $>$ 32)

Variable	BMI $\leq$ 32	BMI $>$ 32	p-value	Type of Test
Final ESS	6.87 $\pm$ 4.57 (6), 0 - 23, N=1012	6.83 $\pm$ 4.49 (6), 0 - 22, N=244	0.9089	Wilcoxon Test
Change in ESS - Baseline to Final	4.59 $\pm$ 5.08 (4), -13 - 23, N=906	4.24 $\pm$ 5.71 (4), -13 - 19, N=224	0.3939	Students t-test

Note:

Format for numeric variables: Mean $\pm$ SD (Median), Range

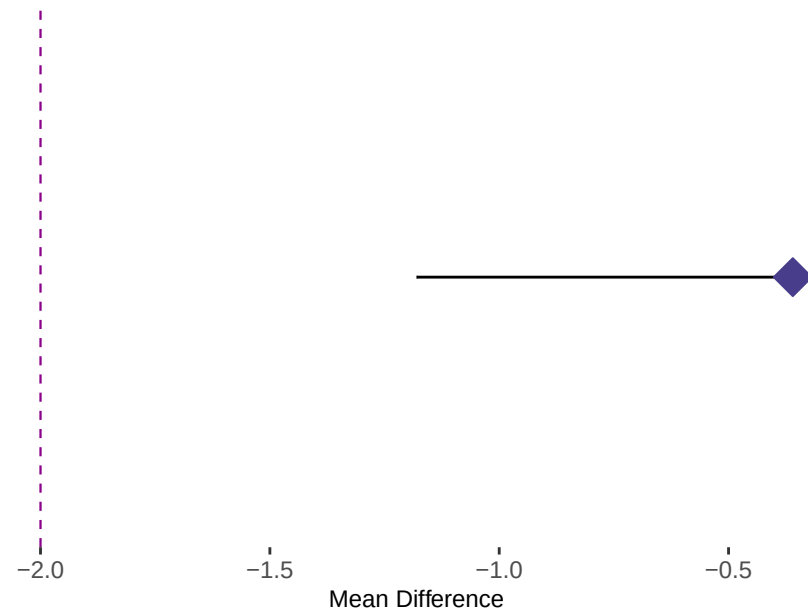
Table 7: ESS Outcomes by Baseline BMI (BMI $\leq$ 32 vs. 32<BMI $\leq$ 35 vs. BMI $>$ 35)

Variable	BMI $\leq$ 32	32<BMI $\leq$ 35	BMI $>$ 35	p-value	Type of Test
Final ESS	6.87 $\pm$ 4.57 (6), 0 - 23 N=1012	6.76 $\pm$ 4.53 (6), 0 - 22 N=186	7.03 $\pm$ 4.4 (7), 0 - 17 N=58	0.8502	Kruskal-Wallis Test
Change in ESS - Baseline to Final	4.59 $\pm$ 5.08 (4), -13 - 23 N=906	4.48 $\pm$ 5.98 (4.5), -13 - 19 N=170	3.48 $\pm$ 4.74 (3), -6 - 17 N=54	0.3117	ANOVA

Note:

Format for numeric variables: Mean $\pm$ SD (Median), Range

There is no statistically significant difference between the BMI groups (BMI $\leq$ 32 vs. BMI $>$ 32 and BMI $\leq$ 32 vs. 32<BMI $\leq$ 35 vs. BMI $>$ 35) in regard to Final ESS and Change in ESS from Baseline to Final Visit. Additionally, as can be seen in the table and graph below, the change in ESS for patients with BMI $>$ 32 is non-inferior to that of patients with BMI $\leq$ 32, as the confidence interval for the difference between the two groups is within the margin of acceptability (ESS change = 2).



The violet dotted lines show the accepted noninferiority margin

Figure 2: Visual of Non-Inferiority analysis for Change in ESS (BMI $\leq$ 32 vs. BMI $>$ 32)

Table 8: Non-inferiority Analysis for Change in ESS from Baseline to Final Follow-up (BMI $\leq$ 32 vs. BMI $>$ 32)

Group	Change in ESS (mean $\pm$ SD)	N	Mean Difference	Lower Bound of Confidence Interval	Accepted Lower Bound	p-value
BMI $\leq$ 32	4.59 $\pm$ 5.08	906	-0.36	-1.18	-2	5e-05
BMI $>$ 32	4.24 $\pm$ 5.71	224				

Usage

Table 9: Usage at Final Visit by Baseline BMI (BMI $\leq$ 32 vs. BMI $>$ 32)

Variable	BMI $\leq$ 32	BMI $>$ 32	p-value	Type of Test
Therapy Use at Final Visit	5.86 $\pm$ 2.2 (6.14), 0 - 10.29, N=937	5.11 $\pm$ 2.29 (5.29), 0 - 10, N=233	1e-05	Students t-test

Note:

Format for numeric variables: Mean $\pm$ SD (Median), Range

Table 10: Usage at Final Visit by Baseline BMI (BMI $\leq$ 32 vs. 32 $<$ BMI $\leq$ 35 vs. BMI $>$ 35)

Variable	BMI $\leq$ 32	32 $<$ BMI $\leq$ 35	BMI $>$ 35	p-value	Type of Test
Therapy Use at Final Visit	5.86 $\pm$ 2.2 (6.14), 0 - 10.29, N=937	5.19 $\pm$ 2.36 (5.29), 0 - 10, N=176	4.89 $\pm$ 2.09 (5.14), 0 - 9.29, N=57	2e-05	ANOVA

Note:

Format for numeric variables: Mean $\pm$ SD (Median), Range

There is a significant difference in the usage (hr/night) between the BMI groups. Patients with BMI $>$ 32 use their device 45 minutes less per night on average than patients with BMI $\leq$ 32.

Clinical Global Impression - Improvement Scale

Table 11: Difference in CGI-I Classification between BMI Subgroups (BMI $\leq$ 32 vs. BMI>32)

CGI-I	BMI $\leq$ 32	BMI>32	p-value	Type of Test
1- Very much improved	N=358 (40.13%)	N=95 (40.6%)	0.387	Fisher's Exact Test (Improved vs Worse/No Change)
2 - Much improved	N=330 (37%)	N=72 (30.77%)		
3 - Minimally improved	N=121 (13.57%)	N=41 (17.52%)		
4 - No change	N=53 (5.94%)	N=18 (7.69%)		
5 - Minimally worse	N=19 (2.13%)	N=5 (2.14%)		
6 - Much worse	N=9 (1.01%)	N=3 (1.28%)		
7 - Very much worse	N=2 (0.22%)	N=0 (0%)		

Table 12: Difference in CGI-I Classification between BMI Subgroups (BMI $\leq$ 32 vs. 32<BMI $\leq$ 35 vs. BMI>35)

CGI-I	BMI $\leq$ 32	32<BMI $\leq$ 35	BMI>35	p-value*
1- Very much improved	N=358 (40.13%)	N=71 (40.57%)	N=24 (40.68%)	0.629
2 - Much improved	N=330 (37%)	N=55 (31.43%)	N=17 (28.81%)	
3 - Minimally improved	N=121 (13.57%)	N=30 (17.14%)	N=11 (18.64%)	
4 - No change	N=53 (5.94%)	N=12 (6.86%)	N=6 (10.17%)	
5 - Minimally worse	N=19 (2.13%)	N=5 (2.86%)	N=0 (0%)	
6 - Much worse	N=9 (1.01%)	N=2 (1.14%)	N=1 (1.69%)	
7 - Very much worse	N=2 (0.22%)	N=0 (0%)	N=0 (0%)	

Note:

*P-value obtained using Fisher's Exact Test (Improved vs Worse/No Change)

Outcomes by Obesity Class

Surgical Success

In the ADHERE Registry 2020 Update, surgical success was defined as a Responder per the Sher 20 criteria (AHI reduction by 50% and final AHI $\leq$ 20). As can be seen in Table 13, the surgical success rate for Obesity Class 1 is 61.1%, 60.9% for Class 2, and 50% for Class 3. The surgical success is not significantly different between Class 1 and Class 2 (p-value = 1).

Table 13: Surgical Success by Obesity Class

Variable	Class 1	Class 2	Class 3	p-value*	Type of Test
Nonresponder - Sher	38.9% (154)	39.1% (18)	50% (4)	1	Fisher's Exact Test
Responder - Sher	61.1% (242)	60.9% (28)	50% (4)	1	

Note:

Class 1 is 30 $\leq$ BMI<35, Class 2 is 35 $\leq$ BMI<40, and Class 3 is BMI $\geq$ 40.

*P-value compares Class 1 and Class 2 due to the small sample size in Class 3.

Therapy Usage

The final therapy usage is not significantly different across the three obesity classes: 5.38 +/- 2.31 hrs/night for Class 1, 5.03 +/- 2.02 hrs/night for Class 2, and 4.17 +/- 2.43 hrs/night for Class 3 (p-value = 0.1879).

Table 14: Usage at Final Visit by Obesity Class

Variable	Class 1	Class 2	Class 3	p-value*	p-value**
Therapy Use at Final Visit	5.38 $\pm$ 2.31 (5.71), 0 - 10.29, N=411	5.03 $\pm$ 2.02 (5.21), 0 - 9, N=48	4.17 $\pm$ 2.43 (3.29), 1.14 - 9.29, N=9	0.2634	0.1879

Note:

Format for numeric variables: Mean $\pm$ SD (Median), Range

Class 1 is 30 $\leq$ BMI<35, Class 2 is 35 $\leq$ BMI<40, and Class 3 is BMI $\geq$ 40.

*P-value compares Class 1 & Class 2 (obtained using a Student's t-test).

**P-value compares all three classes (obtained using ANOVA).

Additional Analysis to Address Deficiency Letter

Logistic Regression - Responder Rate (Sher 20) as Outcome

Logistic Regression Results

Within the deficiency letter, it was suggested that in previous papers, lower BMI was associated with an increased odds of having a more favorable AHI response. The methods used to obtain the results in the Thaler, et al. 2019 publication were performed using the July 2022 ADHERE registry dataset. The tables below show the updated logistic regression results for the Responder Rate outcome, defined by the Sher 20 criteria (AHI $\leq$ 20 and 50% reduction in AHI).

Table 15: Univariate Logistic Regression for Responder Rate (Sher 20) by Baseline BMI

Covariate	Estimate	Std. Error	OR (95% CI)	z value	p-value
(Intercept)	2.054	0.537	7.799 (2.74, 22.53)	3.828	1.3e-04
Baseline BMI	-0.050	0.018	0.951 (0.92, 0.99)	-2.753	5.9e-03

Table 16: Full Logistic Regression Model for Responder Rate (Sher 20)

Covariate	Estimate	Std. Error	OR (95% CI)	z value	p-value
(Intercept)	1.181	0.730	3.257 (0.78, 13.72)	1.618	1.1e-01
Gender - Female	0.494	0.179	1.639 (1.16, 2.34)	2.757	5.8e-03
Final Therapy Usage	0.051	0.032	1.052 (0.99, 1.12)	1.585	1.1e-01
Age	0.003	0.007	1.003 (0.99, 1.02)	0.466	6.4e-01
Baseline BMI	-0.045	0.019	0.956 (0.92, 0.99)	-2.419	1.6e-02
Baseline AHI	0.005	0.005	1.005 (1, 1.01)	0.965	3.3e-01

Table 17: Reduced Logistic Regression for Responder Rate (Sher 20)

Covariate	Estimate	Std. Error	OR (95% CI)	z value	p-value
(Intercept)	1.491	0.604	4.44 (1.37, 14.62)	2.468	1.4e-02
Gender - Female	0.495	0.177	1.641 (1.17, 2.34)	2.801	5.1e-03
Final Therapy Usage	0.051	0.032	1.052 (0.99, 1.12)	1.596	1.1e-01
Baseline BMI	-0.044	0.019	0.957 (0.92, 0.99)	-2.369	1.8e-02

Logistic Regression - AHI<10 and 50 Percent AHI Reduction as Response Outcome (Thaler Publication)

Logistic Regression Results

Within the deficiency letter, it was suggested that in previous papers, “using both definitions, female sex and lower BMI remained as positive predictors of AHI response.” The methods used to obtain the results in the Thaler, et al. 2019 publication were performed using the July 2022 ADHERE registry dataset. The tables below show the updated logistic regression results for the Responder Rate outcome, defined as AHI<10 and 50% reduction in AHI. In contrast to the Thaler, et al. publication, baseline BMI was not a significant predictor in the full multivariate regression model.

Table 18: Univariate Logistic Regression for Responder Rate (AHI<10 and 50 Percent Reduction in AHI) by Baseline BMI

Covariate	Estimate	Std. Error	OR (95% CI)	z value	p-value
(Intercept)	1.206	0.517	3.339 (1.22, 9.27)	2.331	2.0e-02
Baseline BMI	-0.048	0.018	0.953 (0.92, 0.99)	-2.733	6.3e-03

Table 19: Full Logistic Regression Model for Responder Rate (AHI<10 and 50 Percent Reduction in AHI)

Covariate	Estimate	Std. Error	OR (95% CI)	z value	p-value
(Intercept)	1.197	0.713	3.311 (0.82, 13.49)	1.679	9.3e-02
Gender - Female	0.392	0.164	1.48 (1.07, 2.04)	2.387	1.7e-02
Final Therapy Usage	0.044	0.032	1.045 (0.98, 1.11)	1.406	1.6e-01
Age	0.001	0.006	1.001 (0.99, 1.01)	0.161	8.7e-01
Baseline BMI	-0.035	0.018	0.966 (0.93, 1)	-1.905	5.7e-02
Baseline AHI	-0.022	0.005	0.978 (0.97, 0.99)	-4.605	4.1e-06

Table 20: Reduced Logistic Regression Model for Responder Rate (AHI<10 and 50 Percent Reduction in AHI)

Covariate	Estimate	Std. Error	OR (95% CI)	z value	p-value
(Intercept)	1.259	0.601	3.522 (1.09, 11.52)	2.095	3.6e-02
Gender - Female	0.396	0.162	1.486 (1.08, 2.04)	2.441	1.5e-02
Final Therapy Usage	0.045	0.031	1.046 (0.98, 1.11)	1.432	1.5e-01
Baseline BMI	-0.035	0.018	0.966 (0.93, 1)	-1.919	5.5e-02
Baseline AHI	-0.022	0.005	0.978 (0.97, 0.99)	-4.602	4.2e-06

Logistic Regression - AHI<10 as Response Outcome (Thaler Publication)

Logistic Regression Results

Within the deficiency letter, it was suggested that in previous papers, “using both definitions, female sex and lower BMI remained as positive predictors of AHI response.” The methods used to obtain the results in the Thaler, et al. 2019 publication were performed using the July 2022 ADHERE registry dataset. The tables below show the updated logistic regression results for the Responder Rate outcome, defined as AHI<10. In contrast to the Thaler, et al. publication, baseline BMI was not a significant predictor in the full multivariate regression model.

Table 21: Univariate Logistic Regression for Responder Rate (AHI<10) by Baseline BMI

Covariate	Estimate	Std. Error	OR (95% CI)	z value	p-value
(Intercept)	1.183	0.516	3.265 (1.19, 9.05)	2.291	2.2e-02
Baseline BMI	-0.047	0.018	0.954 (0.92, 0.99)	-2.661	7.8e-03

Table 22: Full Logistic Regression Model for Responder Rate (AHI<10)

Covariate	Estimate	Std. Error	OR (95% CI)	z value	p-value
(Intercept)	1.300	0.715	3.668 (0.91, 14.99)	1.819	6.9e-02
Gender - Female	0.452	0.165	1.571 (1.14, 2.17)	2.742	6.1e-03
Final Therapy Usage	0.042	0.032	1.043 (0.98, 1.11)	1.317	1.9e-01
Age	0.000	0.006	1 (0.99, 1.01)	-0.017	9.9e-01
Baseline BMI	-0.033	0.018	0.967 (0.93, 1)	-1.817	6.9e-02
Baseline AHI	-0.024	0.005	0.977 (0.97, 0.99)	-4.924	8.5e-07

Table 23: Reduced Logistic Regression for Responder Rate (AHI<10)

Covariate	Estimate	Std. Error	OR (95% CI)	z value	p-value
(Intercept)	1.638	0.543	5.146 (1.79, 15.05)	3.016	2.6e-03
Gender - Female	0.464	0.162	1.591 (1.16, 2.19)	2.858	4.3e-03
Baseline BMI	-0.037	0.018	0.964 (0.93, 1)	-2.027	4.3e-02
Baseline AHI	-0.024	0.005	0.976 (0.97, 0.99)	-4.990	6.0e-07

Additional Analysis for High BMI Groups

Patients per Group

The following table displays the number of patients in each BMI subgroup ($\text{BMI} \leq 32$, $32 < \text{BMI} \leq 35$, $35 < \text{BMI} \leq 40$, $\text{BMI} > 40$). Patients included had BMI collected at baseline and had a final follow up visit with at least one of the following values recorded at the final visit: AHI, ESS, Therapy Usage, CGI, and/or Patient Satisfaction data.

Table 24: Number of Patients per BMI Subgroup

Variable	$\text{BMI} \leq 32$	$32 < \text{BMI} \leq 35$	$35 < \text{BMI} \leq 40$	$\text{BMI} > 40$
N	1218 (80.8%)	222 (14.7%)	57 (3.8%)	11 (0.7%)

Baseline and Final AHI per Group

Table 25: AHI Outcomes by Baseline BMI Subgroup

Variable	$\text{BMI} \leq 32$	$32 < \text{BMI} \leq 35$	$35 < \text{BMI} \leq 40$	$\text{BMI} > 40$	p-value
Baseline AHI	35.5±14.26 (32.7), 15 - 102.9, N=1218	38.89±16.54 (36.5), 15.5 - 118.7, N=222	39.11±18.67 (36.3), 15.1 - 86, N=57	42.02±15.16 (46.4), 15.6 - 62, N=11	0.0203
Final AHI	15.34±14.44 (10.95), 0 - 83.8, N=930	17.39±16.9 (12.95), 0 - 96.2, N=170	17.01±16.78 (13.35), 0 - 92, N=46	19.8±17.78 (12.65), 5 - 57.9, N=8	0.3167
Change in AHI - Baseline to Final	20.09±18.11 (19.25), -41.6 - 92.3, N=930	21.15±20.13 (19.95), -46.3 - 103.2, N=170	18.43±17.46 (18), -27.5 - 56.2, N=46	20.11±14.25 (25.74), -5.93 - 35.1, N=8	0.8342

Note:

P-value comparing BMI groups derived using a non-parametric Kruskal-Wallis test.

Additional Analysis to Address Deficiency Letter - January 2023

Baseline Demographics by Final Visit Completion

A total of 3603 patients with baseline BMI collected were implanted by the date of data download (7/25/2022). Of these patients, 2337 were expected to have a final visit by 7/25/2022. A total of 1508 (64.53%) completed the final visit. (Patients were considered to have missed the final visit if they were implanted more than 2 years prior to 7/25/2022.)

There were 4 out of the 3603 implanted patients with a baseline BMI that had an unknown implant date. 3 of these patients completed the final follow-up and were thus included in the analysis below.

Table 26: Differences in Baseline Characteristics for patients who followed up vs. those who did not

Variable	Completed Final Visit	Missed Final Visit	p-value*
N	1508 (64.5%)	829 (35.5%)	
Age	60.17±10.61 (61), 21 - 85, N=1502	60.74±11.09 (62), 21 - 87, N=827	0.2204
AHI - Baseline	36.18±14.86 (33.35), 15 - 118.7, N=1508	35.9±15.5 (32.8), 15 - 105.3, N=829	0.6633
BMI - Baseline	29.06±3.76 (29.05), 17.71 - 47.19, N=1508	29.55±3.84 (29.58), 16.95 - 44.92, N=829	0.0026
BMI≤32	80.8% (1218)	75.6% (627)	0.013
32<BMI≤35	14.7% (222)	17.4% (144)	
35<BMI≤40	3.8% (57)	6.2% (51)	
BMI>40	0.7% (11)	0.8% (7)	
ESS - Baseline	11.35±5.54 (11), 0 - 24, N=1326	11.15±5.61 (11), 0 - 24, N=719	0.4479
Female	24% (360)	32.4% (268)	<0.001
White	95.7% (1431)	94.7% (769)	0.3
Hispanic or Latino	2% (29)	2.4% (19)	0.54
Prior Oral App Use	22.6% (339)	23.6% (195)	0.606
Prior CPAP Use	97.9% (1469)	96.7% (799)	0.095

Note:

Format for numeric variables: Mean±SD (Median), Range.

*P-value obtained using a Student's t-test for numeric variables and Fisher's exact test for categorical variables.

The patients that completed the final follow-up had a significantly lower BMI, on average, as compared to those patients that missed the final visit, and the distribution of the percentage of patients across the four BMI categories differed between the completed vs. missed final visit groups (p-values < 0.05). Patients with a baseline BMI ≤ 32 had a follow-up rate of 66%, patients with BMI >32 and ≤ 35 had a follow-up rate of 60.7%, patients with BMI >35 and ≤ 40 had a follow-up rate of 52.8%, and patients with BMI >40 had a follow-up rate of 61.1%.

Comparison of Demographics and Outcomes by BMI Group (BMI ≤ 32 vs. $32 < \text{BMI} \leq 35$ vs. $35 < \text{BMI} \leq 40$ vs. BMI >40)

Demographics and Baseline Characteristics by BMI Group

Table 27: Demographics and Baseline Characteristics by Baseline BMI (BMI ≤ 32 vs. $32 < \text{BMI} \leq 35$ vs. $35 < \text{BMI} \leq 40$ vs. BMI >40)

Variable	BMI ≤ 32	$32 < \text{BMI} \leq 35$	$35 < \text{BMI} \leq 40$	BMI >40	p-value*
N	1218 (80.8%)	222 (14.7%)	57 (3.8%)	11 (0.7%)	
Age	60.34 $\pm$ 10.73 (61), 21 - 85, N=1212	59.94 $\pm$ 10.04 (60), 28 - 81, N=222	57.51 $\pm$ 9.89 (58), 33 - 77, N=57	59 $\pm$ 11.75 (62), 43 - 77, N=11	0.2462
AHI - Baseline	35.5 $\pm$ 14.26 (32.7), 15 - 102.9, N=1218	38.89 $\pm$ 16.54 (36.5), 15.5 - 118.7, N=222	39.11 $\pm$ 18.67 (36.3), 15.1 - 86, N=57	42.02 $\pm$ 15.16 (46.4), 15.6 - 62, N=11	0.0031
BMI - Baseline	27.79 $\pm$ 2.8 (28.18), 17.71 - 32, N=1218	33.33 $\pm$ 0.86 (33.14), 32.01 - 34.97, N=222	36.62 $\pm$ 1.29 (36.3), 35.15 - 39.95, N=57	43.24 $\pm$ 2.79 (42.78), 40.06 - 47.19, N=11	<0.001
ESS - Baseline	11.39 $\pm$ 5.52 (11), 0 - 24, N=1069	11.29 $\pm$ 5.66 (11.5), 0 - 24, N=196	10.69 $\pm$ 5.41 (11), 0 - 22, N=52	12 $\pm$ 5.43 (12), 2 - 21, N=9	0.8181
Female	24.1% (292)	23.5% (52)	25% (14)	18.2% (2)	0.991
White	95.6% (1157)	96.8% (210)	93% (53)	100% (11)	0.58
Hispanic or Latino	2.2% (26)	0.9% (2)	1.8% (1)	0% (0)	0.608
Prior Oral App Use	23.6% (285)	18% (40)	21.1% (12)	18.2% (2)	0.34
Prior CPAP Use	98.1% (1187)	96.8% (215)	98.2% (56)	100% (11)	0.507

Note:

Format for numeric variables: Mean $\pm$ SD (Median), Range.

*P-value comparing the four BMI groups, derived using ANOVA for numeric variables and Fisher's exact test for categorical variables.

Outcomes by BMI Group

Table 28: Outcomes by Baseline BMI (BMI $\leq$ 32 vs. 32<BMI $\leq$ 35 vs. 35<BMI $\leq$ 40 vs. BMI>40)

Variable	BMI $\leq$ 32	32<BMI $\leq$ 35	35<BMI $\leq$ 40	BMI>40	p-value	Type of Test
Final AHI	15.34 $\pm$ 14.44 (10.95), 0 - 83.8, N=930	17.39 $\pm$ 16.9 (12.95), 0 - 96.2, N=170	17.01 $\pm$ 16.78 (13.35), 0 - 92, N=46	19.8 $\pm$ 17.78 (12.65), 5 - 57.9, N=8	0.3167	Kruskal-Wallis Test
Change in AHI - Baseline to Final	20.09 $\pm$ 18.11 (19.25), -41.6 - 92.3, N=930	21.15 $\pm$ 20.13 (19.95), -46.3 - 103.2, N=170	18.43 $\pm$ 17.46 (18), -27.5 - 56.2, N=46	20.11 $\pm$ 14.25 (25.74), -5.93 - 35.1, N=8	0.8242	ANOVA
Responder at Final Visit (Sher)	65.1% (605)	60.6% (103)	60.9% (28)	50% (4)	0.5080	Fisher's Exact Test
Final ESS	6.87 $\pm$ 4.57 (6), 0 - 23, N=1012	6.76 $\pm$ 4.53 (6), 0 - 22, N=186	6.94 $\pm$ 4.59 (6.5), 0 - 17, N=50	7.62 $\pm$ 3.11 (7), 4 - 14, N=8	0.8206	Kruskal-Wallis Test
Change in ESS - Baseline to Final	4.59 $\pm$ 5.08 (4), -13 - 23, N=906	4.48 $\pm$ 5.98 (4.5), -13 - 19, N=170	3.6 $\pm$ 5 (3), -6 - 17, N=47	2.71 $\pm$ 2.43 (3), -2 - 5, N=7	0.4744	ANOVA
Therapy Use (Hr/Night) at Final Visit	5.86 $\pm$ 2.2 (6.14), 0 - 10.29, N=937	5.19 $\pm$ 2.36 (5.29), 0 - 10, N=176	5.03 $\pm$ 2.02 (5.21), 0 - 9, N=48	4.17 $\pm$ 2.43 (3.29), 1.14 - 9.29, N=9	0.0001	ANOVA

Note:

Format for numeric variables: Mean $\pm$ SD (Median), Range

At least one of the BMI groups is significantly different in terms of Final Therapy Usage (hr/night) (p-value < 0.05). There are no significant differences in terms of AHI and ESS outcomes between the BMI groups.

Table 29: Difference in CGI-I Classification between BMI Subgroups (BMI $\leq$ 32 vs. 32<BMI $\leq$ 35 vs. 35<BMI $\leq$ 40 vs. BMI>40)

CGI-I	BMI $\leq$ 32	32<BMI $\leq$ 35	35<BMI $\leq$ 40	BMI>40	p-value*
1- Very much improved	N=358 (40.13%)	N=71 (40.57%)	N=22 (43.14%)	N=2 (25%)	0.361
2 - Much improved	N=330 (37%)	N=55 (31.43%)	N=15 (29.41%)	N=2 (25%)	
3 - Minimally improved	N=121 (13.57%)	N=30 (17.14%)	N=9 (17.65%)	N=2 (25%)	
4 - No change	N=53 (5.94%)	N=12 (6.86%)	N=4 (7.84%)	N=2 (25%)	
5 - Minimally worse	N=19 (2.13%)	N=5 (2.86%)	N=0 (0%)	N=0 (0%)	
6 - Much worse	N=9 (1.01%)	N=2 (1.14%)	N=1 (1.96%)	N=0 (0%)	
7 - Very much worse	N=2 (0.22%)	N=0 (0%)	N=0 (0%)	N=0 (0%)	

Note:

*P-value derived using Fisher's Exact Test (Improved vs. Worse/No Change)

Outcomes for Patients with $32 < \text{BMI} \leq 40$ by Region (OUS vs. US)

Outcomes at the Final visit for patients with a baseline $\text{BMI} > 32$ and $\text{BMI} \leq 40$ were compared by region (OUS vs. US). The tables below show that Final ESS and Therapy Usage were significantly different between regions, with OUS patients having a higher final ESS and lower therapy use on average as compared to US patients (p-values < 0.05).

Table 30: Final Outcomes by Region (OUS vs. US) for Patients with $32 < \text{BMI} \leq 40$ at Baseline

Variable	OUS	US	p-value	Type of Test
Final AHI	15.94±13.89 (12.7), 0 - 92, N=102	18.53±19.07 (13.55), 0 - 96.2, N=114	0.7593	Wilcoxon Test
Change in AHI - Baseline to Final	20.65±17.27 (20.8), -27.5 - 72.7, N=102	20.49±21.52 (18.15), -46.3 - 103.2, N=114	0.9503	Students t-test
Responder at Final Visit (Sher)	61.8% (63)	59.6% (68)	0.7810	Fisher's Exact Test
Final ESS	7.79±4.72 (8), 0 - 22, N=103	6.04±4.24 (5), 0 - 19, N=133	0.0031	Wilcoxon Test
Change in ESS - Baseline to Final	4.08±6.12 (4), -12 - 19, N=99	4.46±5.5 (4), -13 - 16, N=118	0.6367	Students t-test
Therapy Use (Hr/Night) at Final Visit	4.8±2.13 (4.86), 0 - 9, N=101	5.44±2.37 (5.57), 0.14 - 10, N=123	0.0324	Students t-test

Note:

Format for numeric variables: Mean±SD (Median), Range

Outcomes for Patients with $32 < \text{BMI} \leq 40$ by Site

Table 31: Final Outcomes by Site for Patients with $32 < \text{BMI} \leq 40$ at Baseline

Site Name	N	Final AHI	Change in AHI - Baseline to Final	Final Sher Response	Final ESS	Change in ESS - Baseline to Final	Final Therapy Use (Hr/Night)
Advanced ENT and Allergy	4	5.15±0.35 (5.15), 4.9 - 5.4, N=2	35.45±33.45 (35.45), 11.8 - 59.1, N=2	100% (2)	6.67±6.11 (8), 0 - 12, N=3	9±9.64 (13), -2 - 16, N=3	6.29±NA (6.29), 6.29 - 6.29, N=1
Asklepios Kliniken Hamburg	1	N=0	N=0	N=0	6±NA (6), 6 - 6, N=1	6±NA (6), 6 - 6, N=1	6±NA (6), 6 - 6, N=1
Cincinnati VA	16	10.64±10.12 (6.75), 0 - 28.3, N=8	33.34±14.3 (32.3), 16.8 - 60.3, N=8	75% (6)	5.69±3.07 (5.5), 1 - 11, N=16	3.75±5.03 (2.5), -6 - 12, N=16	5.83±1.54 (5.86), 2.86 - 8.14, N=15
Cleveland Clinic Foundation	8	17.73±11.28 (15.4), 2.9 - 38.3, N=7	23.74±17.76 (25.5), -3.5 - 42.3, N=7	57.1% (4)	9.43±5.94 (10), 2 - 16, N=7	4.57±7.28 (2), -2 - 16, N=7	4.09±2.04 (4.5), 0.86 - 6.57, N=8
Hackensack University Medical Center	4	41.97±39.99 (22.1), 15.8 - 88, N=3	0.53±24.6 (0.4), -24 - 25.2, N=3	33.3% (1)	6.25±5.12 (6), 1 - 12, N=4	-1.5±2.65 (-1), -5 - 1, N=4	3.29±3.4 (2.43), 0.43 - 7.86, N=4
Indiana University	8	23.6±17.86 (17), 7.8 - 55.1, N=7	8.34±11.39 (7.9), -7.2 - 22.8, N=7	42.9% (3)	13±5.66 (13), 9 - 17, N=2	-1±0 (-1), -1 - -1, N=2	6.57±1.14 (6.57), 5.43 - 7.71, N=3
Kansas University Medical Center	5	23.47±9.27 (22.7), 14.6 - 33.1, N=3	7.37±13.77 (11.9), -8.1 - 18.3, N=3	0% (0)	1±1.41 (1), 0 - 2, N=2	2±1.41 (2), 1 - 3, N=2	4.57±2.85 (5), 0.86 - 7.43, N=4
Kantonsspital Baselland	3	18.55±13.9 (12.1), 9.05 - 34.5, N=3	15.02±27.65 (3.6), -5.1 - 46.55, N=3	33.3% (1)	10.5±2.12 (10.5), 9 - 12, N=2	-3.5±7.78 (-3.5), -9 - 2, N=2	7±1.41 (7), 6 - 8, N=2
Klinik für HNO-Heilkunde/HNO-Schlaf Labor	32	19.56±16.64 (18), 1.6 - 92, N=31	13.64±15.37 (15), -27.5 - 45, N=31	48.4% (15)	7.75±4.59 (7), 1 - 19, N=32	6.03±6.04 (7), -12 - 17, N=31	5.35±2.01 (5.43), 0.71 - 8.86, N=31
Medical College of Wisconsin	6	20.15±15.49 (13.95), 10 - 51, N=6	23.68±17.48 (19), 2 - 53.1, N=6	66.7% (4)	7.83±3.54 (8), 4 - 13, N=6	6±4.05 (6), 0 - 12, N=6	4.57±2.25 (3.79), 2.86 - 7.86, N=4
MetroHealth	4	38.6±NA (38.6), 38.6 - 38.6, N=1	-10.8±NA (-10.8), -10.8 - -10.8, N=1	0% (0)	7.5±4.95 (7.5), 4 - 11, N=2	1.5±2.12 (1.5), 0 - 3, N=2	5.57±NA (5.57), 5.57 - 5.57, N=1

Table 31: Final Outcomes by Site for Patients with $32 < \text{BMI} \leq 40$
at Baseline (*continued*)

Site Name	N	Final AHI	Change in AHI - Baseline to Final	Final Sher Response	Final ESS	Change in ESS - Baseline to Final	Final Therapy Use (Hr/Night)
Middlesex Hospital	1	N=0	N=0	N=0	3±NA (3), 3 - 3, N=1	1±NA (1), 1 - 1, N=1	4.71±NA (4.71), 4.71 - 4.71, N=1
Northshore University	2	N=0	N=0	N=0	2.5±0.71 (2.5), 2 - 3, N=2	N=0	6±0 (6), 6 - 6, N=2
Ogden Clinic	3	5.33±0.49 (5.1), 5 - 5.9, N=3	23.7±18.55 (13.9), 12.1 - 45.1, N=3	100% (3)	5±4.58 (4), 1 - 10, N=3	2.67±2.31 (4), 0 - 4, N=3	9.05±0.22 (9), 8.86 - 9.29, N=3
OLVG	3	13.27±10.95 (18.3), 0.7 - 20.8, N=3	26.53±17.17 (22.7), 11.6 - 45.3, N=3	66.7% (2)	9.67±2.89 (8), 8 - 13, N=3	2.67±1.53 (3), 1 - 4, N=3	4.29±2.81 (5.14), 1.14 - 6.57, N=3
Pennsylvania Hospital	3	9.8±11.6 (9.8), 1.6 - 18, N=2	30.55±5.02 (30.55), 27 - 34.1, N=2	100% (2)	N=0	N=0	7.57±NA (7.57), 7.57 - 7.57, N=1
Puget Sound ENT	1	N=0	N=0	N=0	N=0	N=0	1.71±NA (1.71), 1.71 - 1.71, N=1
Reading Hospital	1	N=0	N=0	N=0	1±NA (1), 1 - 1, N=1	N=0	N=0
Richard L Roudebush VA Medical Center	3	34.6±NA (34.6), 34.6 - 34.6, N=1	-3.3±NA (-3.3), -3.3 - -3.3, N=1	0% (0)	8.33±9.45 (5), 1 - 19, N=3	3±6.08 (0), -1 - 10, N=3	1.62±2.2 (0.57), 0.14 - 4.14, N=3
Rush University Medical Center	2	4±NA (4), 4 - 4, N=1	19.1±NA (19.1), 19.1 - 19.1, N=1	100% (1)	7±5.66 (7), 3 - 11, N=2	2±NA (2), 2 - 2, N=1	7.14±1.21 (7.14), 6.29 - 8, N=2
Sleep Management Institute	1	2.8±NA (2.8), 2.8 - 2.8, N=1	16.2±NA (16.2), 16.2 - 16.2, N=1	100% (1)	6±NA (6), 6 - 6, N=1	N=0	4.57±NA (4.57), 4.57 - 4.57, N=1
St. Cloud Ear, Nose & Throat	3	1.5±NA (1.5), 1.5 - 1.5, N=1	22.3±NA (22.3), 22.3 - 22.3, N=1	100% (1)	6±0 (6), 6 - 6, N=2	8.5±3.54 (8.5), 6 - 11, N=2	N=0
St. Joseph Mercy Hospital	1	56.3±NA (56.3), 56.3 - 56.3, N=1	-13.2±NA (-13.2), -13.2 - -13.2, N=1	0% (0)	7±NA (7), 7 - 7, N=1	5±NA (5), 5 - 5, N=1	6.71±NA (6.71), 6.71 - 6.71, N=1
St. Luke Presbyterian Hospital	3	7.27±7.25 (3.8), 2.4 - 15.6, N=3	20.6±14.64 (17.9), 7.5 - 36.4, N=3	66.7% (2)	3±NA (3), 3 - 3, N=1	6±NA (6), 6 - 6, N=1	3.9±2.14 (4), 1.71 - 6, N=3
Technische Universitat Dresden	2	6.75±0.35 (6.75), 6.5 - 7, N=2	44.4±40.02 (44.4), 16.1 - 72.7, N=2	100% (2)	4.5±0.71 (4.5), 4 - 5, N=2	7.5±9.19 (7.5), 1 - 14, N=2	6.71±2.83 (6.71), 4.71 - 8.71, N=2

Table 31: Final Outcomes by Site for Patients with $32 < \text{BMI} \leq 40$
at Baseline (*continued*)

Site Name	N	Final AHI	Change in AHI - Baseline to Final	Final Sher Response	Final ESS	Change in ESS - Baseline to Final	Final Therapy Use (Hr/Night)
Technischen Universität München	38	11.91±11.19 (9.1), 0.3 - 45.8, N=37	26.78±15.15 (26.8), -3.2 - 61.2, N=37	78.4% (29)	8.31±5.02 (8.5), 0 - 20, N=36	3.45±6.04 (4), -5 - 19, N=33	4.56±2.14 (4.43), 0 - 9, N=37
The Ohio State University	8	4.07±2.42 (4.1), 0.5 - 7.4, N=6	34.45±19.5 (26.3), 17.7 - 67.3, N=6	100% (6)	4.17±4.4 (3), 0 - 12, N=6	6.83±4.49 (8), 0 - 12, N=6	5.4±1.61 (5.29), 3.71 - 8.29, N=6
The University of Iowa	1	N=0	N=0	N=0	15±NA (15), 15 - 15, N=1	-11±NA (-11), -11 - -11, N=1	N=0
The University of Texas Southwestern Medical Center	7	13.1±12.75 (8.35), 4 - 31.7, N=4	26.27±20.12 (24.85), 5.2 - 50.19, N=4	75% (3)	4±1.79 (3.5), 2 - 7, N=6	3.83±3.54 (4), 0 - 9, N=6	4.86±3.52 (5.29), 1.14 - 8.14, N=3
Thomas Jefferson University and Hospital	14	15.76±11.22 (14.2), 3.1 - 38.6, N=8	20.57±20.28 (14.9), 2.15 - 66.9, N=8	62.5% (5)	6.1±4.04 (5.5), 1 - 13, N=10	2.5±3.39 (2.5), -2 - 7, N=6	5.39±2.8 (4.07), 2.43 - 10, N=10
Universitäts-HNO-Klinik Mannheim	28	18.59±13.74 (16), 0 - 49.8, N=25	17.56±16.02 (19.6), -16.5 - 41.9, N=25	52% (13)	7.04±4.95 (7), 1 - 22, N=27	3±6.04 (1), -6 - 17, N=27	4.27±1.99 (4.57), 0 - 8.43, N=24
Universitätsklinikum Hamburg Eppendorf	1	5.1±NA (5.1), 5.1 - 5.1, N=1	40.9±NA (40.9), 40.9 - 40.9, N=1	100% (1)	N=0	N=0	1.14±NA (1.14), 1.14 - 1.14, N=1
University of Alabama at Birmingham	13	20.69±8.99 (22.3), 6.6 - 32.8, N=8	10.91±11 (11.93), -2.5 - 27, N=8	25% (2)	5.18±2.32 (5), 2 - 8, N=11	7±5.74 (9), -2 - 15, N=9	5.17±2.43 (5.57), 0.86 - 8.29, N=9
University of Colorado Denver	3	N=0	N=0	N=0	6±7.81 (2), 1 - 15, N=3	0±3 (0), -3 - 3, N=3	6.86±NA (6.86), 6.86 - 6.86, N=1
University of Pennsylvania	7	47.7±31.32 (44), 10.3 - 96.2, N=5	13.82±37.16 (18.8), -46.3 - 56.2, N=5	20% (1)	6.5±3.78 (5.5), 2 - 12, N=6	8.8±5.45 (7), 4 - 16, N=5	5.18±3.16 (5.79), 0.86 - 8.29, N=4
University of Pittsburgh Medical Center	10	8.52±6.51 (9), 0 - 15.7, N=6	25.8±16.47 (24.05), 5.5 - 45.2, N=6	66.7% (4)	5.9±3.31 (5.5), 1 - 12, N=10	6±4.76 (7), -2 - 12, N=10	7.03±2.14 (7.57), 1.71 - 8.71, N=9

Table 31: Final Outcomes by Site for Patients with $32 < \text{BMI} \leq 40$
at Baseline (*continued*)

Site Name	N	Final AHI	Change in AHI - Baseline to Final	Final Sher Response	Final ESS	Change in ESS - Baseline to Final	Final Therapy Use (Hr/Night)
University of South Florida	3	46.15±52.96 (46.15), 8.7 - 83.6, N=2	-17.25±36.7 (-17.25), -43.2 - 8.7, N=2	50% (1)	13.5±0.71 (13.5), 13 - 14, N=2	-5±NA (-5), -5 - -5, N=1	N=0
University of Southern California	2	22.05±10.54 (22.05), 14.6 - 29.5, N=2	12.85±0.92 (12.85), 12.2 - 13.5, N=2	0% (0)	14±NA (14), 14 - 14, N=1	3±NA (3), 3 - 3, N=1	5.86±4.44 (5.86), 2.71 - 9, N=2
University of Tennessee Health Science Center Valley ENT	1 4	N=0 23.44±15.69 (24.4), 5.5 - 39.45, N=4	N=0 21.44±11.98 (21.02), 7.2 - 36.5, N=4	N=0 50% (2)	2±NA (2), 2 - 2, N=1 7.67±7.23 (4), 3 - 16, N=3	9±NA (9), 9 - 9, N=1 -2.5±14.85 (-2.5), -13 - 8, N=2	8±NA (8), 8 - 8, N=1 4.33±3.32 (5.57), 0.57 - 6.86, N=3
Vanderbilt University Medical Center	4	9.65±5.89 (10.6), 1.9 - 15.5, N=4	41.43±42.18 (24.35), 13.8 - 103.2, N=4	100% (4)	3.5±0.71 (3.5), 3 - 4, N=2	6.5±10.61 (6.5), -1 - 14, N=2	5.48±2.5 (5.43), 3 - 8, N=3
Virginia Commonwealth University	2	4.45±0.64 (4.45), 4 - 4.9, N=2	42.05±2.9 (42.05), 40 - 44.1, N=2	100% (2)	2.5±3.54 (2.5), 0 - 5, N=2	8±NA (8), 8 - 8, N=1	7.57±3.43 (7.57), 5.14 - 10, N=2
Weill Cornell Medicine	13	21.53±25.09 (14), 2.4 - 83.4, N=13	20.72±22.98 (20.3), -36.5 - 50.7, N=13	61.5% (8)	4.4±1.58 (4), 2 - 6, N=10	6.6±3.24 (6), 2 - 11, N=10	5.87±1.55 (5.36), 4.14 - 9.86, N=12

Note:

Format for numeric variables: Mean±SD (Median), Range

N=0 indicates that no patients at this site had data entered for this variable.



Label Expansion - Adverse Events

ADHERE Registry

Lauren Makey

02/14/23

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Implant Adverse Events by Baseline AHI

Patients that were enrolled in the ADHERE registry by July 22, 2022, had data through implant collected, and had AHI collected at baseline were included in the implant adverse event analysis.

Implant Adverse Events by Seriousness

Table 1: Implant Adverse Events by Seriousness and Baseline AHI

Implant AEs	AHI</=65 (N = 3572)		65<AHI</=75 (N = 65)		75<AHI</=100 (N = 58)		AHI>100 (N = 4)		p-value*
	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events	
Total	151	131 (3.7%)	1	1 (1.5%)	3	2 (3.4%)	1	1 (25%)	
Serious	24	24 (0.7%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0.2
Non-Serious	125	105 (2.9%)	1	1 (1.5%)	3	2 (3.4%)	1	1 (25%)	
Unrelated	2	2 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	

Note:

Format for Subjects with Events: Number of subjects with at least one event (% of subjects per AHI group)

*p-value derived from Fisher’s exact test excluding unrelated adverse events

Per the table above, there was no significant difference in the percentage of patients with at least one implant event (serious or non-serious) between the AHI groups (p-value = 0.2).

Serious Implant Adverse Events by Type

Table 2: Serious Implant Adverse Events by Type and Baseline AHI

Serious AEs	AHI < /=65 (N = 3572)		65 < AHI < /=75 (N = 65)		75 < AHI < /=100 (N = 58)		AHI > 100 (N = 4)	
	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events
Total	24	24 (0.7%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Revision - Sense Lead	6	6 (0.2%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Revision - IPG	4	4 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Tachycardia	2	2 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Pneumothorax	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Infection	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Hematoma	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Hypotension	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Rhabdomyolysis	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Intraoperative arrest/bradycardia	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Cervical Swelling with submandibular Hematoma	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Revision - Stim Lead	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Revision - Stimulation Lead, IPG	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Chest pain with Tachycardia	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Bradycardia	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Intraoperative revision - IPG	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)

Note:

Format for Subjects with Events: Number of subjects with at least one event (% of subjects per AHI group)

Non-Serious Implant Adverse Events by Type

Table 3: Non-Serious Implant Adverse Events by Type and Baseline AHI

Non-Serious AEs	AHI</=65 (N = 3572)			65<AHI</=75 (N = 65)			75<AHI</=100 (N = 58)			AHI>100 (N = 4)		
	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events
Total	125	105 (2.9%)	1	1 (1.5%)	3	2 (3.4%)	1	1 (25%)	1	1 (25%)	1	1 (25%)
Hematoma	14	14 (0.4%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Intraoperative bleeding	10	10 (0.3%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Speech Difficulties	7	7 (0.2%)	0	0 (0%)	1	1 (1.7%)	1	1 (1.7%)	0	0 (0%)	0	0 (0%)
Headache	6	6 (0.2%)	0	0 (0%)	1	1 (1.7%)	1	1 (1.7%)	0	0 (0%)	0	0 (0%)
Tongue weakness	6	6 (0.2%)	1	1 (1.5%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Incision discomfort/irritation	5	5 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Pneumothorax	5	5 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Neuropaxia	5	5 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Tongue Discomfort/irritation	4	4 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Unknown	4	4 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Chest pain	4	4 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Seroma	4	4 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Infection	3	3 (0.1%)	0	0 (0%)	1	1 (1.7%)	1	1 (25%)	1	1 (25%)	1	1 (25%)
Lip weakness	3	3 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Postoperative bleeding	3	3 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Ecchymosis	3	3 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Edema	3	3 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Facial Swelling	3	3 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Nerve weakness	2	2 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Urinary retention	2	2 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Tongue Deviation	2	2 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Sore throat	2	2 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)

Table 3: Non-Serious Implant Adverse Events by Type and Baseline AHI (*continued*)

Non-Serious AEs	AHI ≤ 65 (N = 3572)		65 < AHI ≤ 75 (N = 65)		75 < AHI ≤ 100 (N = 58)		AHI > 100 (N = 4)	
	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events
Postoperative desaturation	2	2 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Atrial fibrillation	2	2 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Intraoperative nerve repositioning	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Cuff placement challenges	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Discomfort - swallowing	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Allergic reaction at incision site	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Tongue movement change	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Neck pain	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Difficult dissection	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
IPG Reposition	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Conversion Disorder	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Cervical Swelling	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Eye pain	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Technical Challenges	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Dysphagia	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Wound Dehiscence	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Discomfort - IPG	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Neck swelling	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Bradycardia	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Incision swelling	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Mouth pain	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Nerve Damage	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Hypotension	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)

Table 3: Non-Serious Implant Adverse Events by Type and Baseline AHI (*continued*)

Non-Serious AEs	AHI</=65 (N = 3572)		65<AHI</=75 (N = 65)		75<AHI</=100 (N = 58)		AHI>100 (N = 4)	
	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events

Note:

Format for Subjects with Events: Number of subjects with at least one event (% of subjects per AHI group)
Subjects could have had more than one implant adverse event

Revisions - Implant Adverse Events

Table 4: Revisions - Implant Adverse Events by Baseline AHI

Revision	AHI</=65 (N = 3572)		65<AHI</=75 (N = 65)		75<AHI</=100 (N = 58)		AHI>100 (N = 4)	
	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events
Total	13	13 (0.4%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Sense Lead	6	6 (0.2%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
IPG	5	5 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Stim Lead	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Stimulation Lead, IPG	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)

Note:

Format for Subjects with Events: Number of subjects with at least one event (% of subjects per AHI group)

Follow-up Adverse Events by Baseline AHI

Patients that were enrolled in the ADHERE registry by July 22, 2022, had AHI collected at baseline, and had a final follow-up visit with at least one of the following values recorded at the final visit - AHI, ESS, Therapy Usage, CGI, and/or Patient Satisfaction - were included in the follow-up adverse event analysis.

Follow-up Adverse Events by Seriousness

Table 5: Follow-up Adverse Events by Seriousness and Baseline AHI

Follow-up AEs	AHI</=65 (N = 1483)		65<AHI</=75 (N = 31)		75<AHI</=100 (N = 26)		AHI>100 (N = 2)		p-value*
	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events	
Total	550	432 (29.1%)	18	13 (41.9%)	13	8 (30.8%)	1	1 (50%)	
Serious	44	42 (2.8%)	1	1 (3.2%)	0	0 (0%)	0	0 (0%)	0.31
Non-Serious	503	399 (26.9%)	17	12 (38.7%)	13	8 (30.8%)	1	1 (50%)	
Unrelated	3	3 (0.2%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	

Note:

Format for Subjects with Events: Number of subjects with at least one event (% of subjects per AHI group)

Subjects could have had more than one follow-up event

*p-value derived from Fisher’s exact test excluding unrelated adverse events

Per the table above, there was no significant difference in the percentage of patients with at least one follow-up event (serious or non-serious) between the AHI groups (p-value = 0.31).

Serious Follow-up Adverse Events by Type

Table 6: Serious Follow-up Adverse Events by Type and Baseline AHI

Serious AEs	AHI < /= 65 (N = 1483)		65 < AHI < /= 75 (N = 31)		75 < AHI < /= 100 (N = 26)		AHI > 100 (N = 2)	
	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events
Total	44	42 (2.8%)	1	1 (3.2%)	0	0 (0%)	0	0 (0%)
Sense Lead Revision	22	21 (1.4%)	1	1 (3.2%)	0	0 (0%)	0	0 (0%)
Stim Lead Revision	7	7 (0.5%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
IPG Re-anchored	3	3 (0.2%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Stim Lead Reattached	3	3 (0.2%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
IPG Revision	3	3 (0.2%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Sense Lead Reattached	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
IPG and Sense Lead Revision	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Swallowing, chewing and talking not possible immediately after surgery	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Sore tongue, difficulty talking while Inspire is off and attacking pain in left lower jaw	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Unspecified	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
System Explant	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)

Note:

Format for Subjects with Events: Number of subjects with at least one event (% of subjects per AHI group)

Subjects could have had more than one follow-up event

Revisions - Follow-up Adverse Events

Table 7: Revisions - Follow-up Adverse Events by Baseline AHI

Revision	AHI < / = 65 (N = 1483)		65 < AHI < / = 75 (N = 31)		75 < AHI < / = 100 (N = 26)		AHI > 100 (N = 2)	
	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events
Total	42	40 (2.7%)	1	1 (3.2%)	0	0 (0%)	0	0 (0%)
Sense Lead Revision	22	21 (1.4%)	1	1 (3.2%)	0	0 (0%)	0	0 (0%)
Stim Lead Revision	7	7 (0.5%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
IPG Re-anchored	3	3 (0.2%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Stim Lead Reattached	3	3 (0.2%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
IPG Revision	3	3 (0.2%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Sense Lead Reattached	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
IPG and Sense Lead Revision	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Unspecified	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
System Explant	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)

Note:

Format for Subjects with Events: Number of subjects with at least one event (% of subjects per AHI group)

Subjects could have had more than one follow-up event

Implant Adverse Events by Baseline BMI

Patients that were enrolled in the ADHERE registry by July 22, 2022, had data through implant collected, and had BMI collected at baseline were included in the implant adverse event analysis.

Implant Adverse Events by Seriousness

Table 8: Implant Adverse Events by Seriousness and Baseline BMI

Implant AEs	BMI < / = 32 (N = 2832)		32 < BMI < / = 35 (N = 592)		35 < BMI < / = 40 (N = 156)		BMI > 40 (N = 23)		p-value*
	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events	
Total	122	105 (3.7%)	21	19 (3.2%)	5	5 (3.2%)	7	5 (21.7%)	
Serious	19	19 (0.7%)	3	3 (0.5%)	2	2 (1.3%)	0	0 (0%)	
Non-Serious	101	84 (3%)	18	16 (2.7%)	3	3 (1.9%)	7	5 (21.7%)	5.7e-03
Unrelated	2	2 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	

Note:

Format for Subjects with Events: Number of subjects with at least one event (% of subjects per BMI group)

Subjects could have had more than one implant adverse event

*p-value derived from Fisher's exact test excluding unrelated adverse events

Per the table above, at least one BMI group had a significantly different percentage of patients with implant event(s) (serious or non-serious) (p-value < 0.01). However, all adverse events in patients with BMI > 40 were non-serious.

Serious Implant Adverse Events by Type

Table 9: Serious Implant Adverse Events by Type and Baseline BMI

Serious AEs	BMI < / = 32 (N = 2832)		32 < BMI < / = 35 (N = 592)		35 < BMI < / = 40 (N = 156)		BMI > 40 (N = 23)	
	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events
Total	19	19 (0.7%)	3	3 (0.5%)	2	2 (1.3%)	0	0 (0%)
Revision - Sense Lead	5	5 (0.2%)	1	1 (0.2%)	0	0 (0%)	0	0 (0%)
Revision - IPG	4	4 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Tachycardia	2	2 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Pneumothorax	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Hypotension	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Intraoperative arrest/bradycardia	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Cervical Swelling with submandibular Hematoma	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Revision - Stim Lead	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Revision - Stimulation Lead, IPG	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Bradycardia	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Intraoperative revision - IPG	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Infection	0	0 (0%)	0	0 (0%)	1	1 (0.6%)	0	0 (0%)
Hematoma	0	0 (0%)	1	1 (0.2%)	0	0 (0%)	0	0 (0%)
Rhabdomyolysis	0	0 (0%)	0	0 (0%)	1	1 (0.6%)	0	0 (0%)
Chest pain with Tachycardia	0	0 (0%)	1	1 (0.2%)	0	0 (0%)	0	0 (0%)

Note:

Format for Subjects with Events: Number of subjects with at least one event (% of subjects per BMI group)

Non-Serious Implant Adverse Events by Type

Table 10: Non-Serious Implant Adverse Events by Type and Baseline BMI

Non-Serious AEs	BMI < /= 32 (N = 2832)		32 < BMI < /= 35 (N = 592)		35 < BMI < /= 40 (N = 156)		BMI > 40 (N = 23)	
	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events
Total	101	84 (3%)	18	16 (2.7%)	3	3 (1.9%)	7	5 (21.7%)
Hematoma	12	12 (0.4%)	2	2 (0.3%)	0	0 (0%)	0	0 (0%)
Intraoperative bleeding	9	9 (0.3%)	1	1 (0.2%)	0	0 (0%)	0	0 (0%)
Speech Difficulties	8	8 (0.3%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Headache	6	6 (0.2%)	1	1 (0.2%)	0	0 (0%)	0	0 (0%)
Tongue weakness	6	6 (0.2%)	1	1 (0.2%)	0	0 (0%)	0	0 (0%)
Pneumothorax	5	5 (0.2%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Infection	4	4 (0.1%)	1	1 (0.2%)	0	0 (0%)	0	0 (0%)
Incision discomfort/irritation	4	4 (0.1%)	1	1 (0.2%)	0	0 (0%)	0	0 (0%)
Neuropaxia	4	4 (0.1%)	1	1 (0.2%)	0	0 (0%)	0	0 (0%)
Postoperative bleeding	3	3 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Seroma	3	3 (0.1%)	1	1 (0.2%)	0	0 (0%)	0	0 (0%)
Tongue	2	2 (0.1%)	1	1 (0.2%)	1	1 (0.6%)	0	0 (0%)
Discomfort/irritation								
Unknown	2	2 (0.1%)	1	1 (0.2%)	1	1 (0.6%)	0	0 (0%)
Nerve weakness	2	2 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Chest pain	2	2 (0.1%)	0	0 (0%)	0	0 (0%)	1	1 (4.3%)
Ecchymosis	2	2 (0.1%)	0	0 (0%)	0	0 (0%)	1	1 (4.3%)
Sore throat	2	2 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Atrial fibrillation	2	2 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Lip weakness	1	1 (0%)	1	1 (0.2%)	0	0 (0%)	1	1 (4.3%)
Cuff placement challenges	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Discomfort - swallowing	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Allergic reaction at incision site	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)

Table 10: Non-Serious Implant Adverse Events by Type and Base-line BMI (*continued*)

Non-Serious AEs	BMI < / = 32 (N = 2832)		32 < BMI < / = 35 (N = 592)		35 < BMI < / = 40 (N = 156)		BMI > 40 (N = 23)	
	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events
Tongue movement change	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Neck pain	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Urinary retention	1	1 (0%)	0	0 (0%)	0	0 (0%)	1	1 (4.3%)
IPG Reposition	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Conversion Disorder	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Cervical Swelling	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Tongue Deviation	1	1 (0%)	1	1 (0.2%)	0	0 (0%)	0	0 (0%)
Edema	1	1 (0%)	0	0 (0%)	1	1 (0.6%)	1	1 (4.3%)
Facial Swelling	1	1 (0%)	1	1 (0.2%)	0	0 (0%)	1	1 (4.3%)
Eye pain	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Postoperative desaturation	1	1 (0%)	1	1 (0.2%)	0	0 (0%)	0	0 (0%)
Technical Challenges	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Discomfort - IPG	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Neck swelling	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Bradycardia	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Incision swelling	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Mouth pain	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Nerve Damage	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Hypotension	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Intraoperative nerve repositioning	0	0 (0%)	1	1 (0.2%)	0	0 (0%)	0	0 (0%)
Difficult dissection	0	0 (0%)	0	0 (0%)	0	0 (0%)	1	1 (4.3%)
Dysphagia	0	0 (0%)	1	1 (0.2%)	0	0 (0%)	0	0 (0%)
Wound Deliscence	0	0 (0%)	1	1 (0.2%)	0	0 (0%)	0	0 (0%)

Note:

Format for Subjects with Events: Number of subjects with at least one event (% of subjects per BMI group)

Subjects could have had more than one implant adverse event

Revisions - Implant Adverse Events

Table 11: Revisions - Implant Adverse Events by Baseline BMI

Revision	BMI</=32 (N = 2832)		32<BMI</=35 (N = 592)		35<BMI</=40 (N = 156)		BMI>40 (N = 23)	
	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events
Total	12	12 (0.4%)	1	1 (0.2%)	0	0 (0%)	0	0 (0%)
Sense Lead	5	5 (0.2%)	1	1 (0.2%)	0	0 (0%)	0	0 (0%)
IPG	5	5 (0.2%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Stim Lead	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Stimulation Lead, IPG	1	1 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)

Note:

Format for Subjects with Events: Number of subjects with at least one event (% of subjects per BMI group)

Follow-up Adverse Events by Baseline BMI

Patients that were enrolled in the ADHERE registry by July 22, 2022, had BMI collected at baseline, and had a final follow-up visit with at least one of the following values recorded at the final visit - AHI, ESS, Therapy Usage, CGI, and/or Patient Satisfaction - were included in the follow-up adverse event analysis.

Follow-up Adverse Events by Seriousness

Table 12: Follow-up Adverse Events by Seriousness and Baseline BMI

Follow-up AEs	BMI < / = 32 (N = 1218)			32 < BMI < / = 35 (N = 222)			35 < BMI < / = 40 (N = 57)			BMI > 40 (N = 11)		
	Total Events	Subjects with Events		Total Events	Subjects with Events		Total Events	Subjects with Events		Total Events	Subjects with Events	p-value*
Total	456	351 (28.8%)		86	71 (32%)		25	20 (35.1%)		1	1 (9.1%)	
Serious	33	32 (2.6%)		8	8 (3.6%)		2	1 (1.8%)		1	1 (9.1%)	
Non-Serious	422	328 (26.9%)		76	63 (28.4%)		23	19 (33.3%)		0	0 (0%)	0.33
Unrelated	1	1 (0.1%)		2	2 (0.9%)		0	0 (0%)		0	0 (0%)	

Note:

Format for Subjects with Events: Number of subjects with at least one event (% of subjects per BMI group)

Subjects could have had more than one follow-up event

*p-value derived from Fisher's exact test excluding unrelated adverse events

Per the table above, there was no significant difference in the percentage of patients with at least one follow-up event (serious or non-serious) between the BMI groups (p-value = 0.33).

Serious Follow-up Adverse Events by Type

Table 13: Serious Follow-up Adverse Events by Type and Baseline BMI

Serious AEs	BMI < / = 32 (N = 1218)		32 < BMI < / = 35 (N = 222)		35 < BMI < / = 40 (N = 57)		BMI > 40 (N = 11)	
	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events
Total	33	32 (2.6%)	8	8 (3.6%)	2	1 (1.8%)	1	1 (9.1%)
Sense Lead Revision	16	15 (1.2%)	4	4 (1.8%)	1	1 (1.8%)	1	1 (9.1%)
Stim Lead Revision	5	5 (0.4%)	2	2 (0.9%)	0	0 (0%)	0	0 (0%)
IPG Re-anchored	3	3 (0.2%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
IPG Revision	3	3 (0.2%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Sense Lead Reattached	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Stim Lead Reattached	1	1 (0.1%)	1	1 (0.5%)	1	1 (1.8%)	0	0 (0%)
IPG and Sense Lead Revision	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Swallowing, chewing and talking not possible immediately after surgery	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Sore tongue, difficulty talking while Inspire is off and attacking pain in left lower jaw	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
System Explant	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Unspecified	0	0 (0%)	1	1 (0.5%)	0	0 (0%)	0	0 (0%)

Note:

Format for Subjects with Events: Number of subjects with at least one event (% of subjects per BMI group)

Subjects could have had more than one follow-up event

Revisions - Follow-up Adverse Events

Table 14: Revisions - Follow-up Adverse Events by Baseline BMI

Revision	BMI</=32 (N = 1218)		32<BMI</=35 (N = 222)		35<BMI</=40 (N = 57)		BMI>40 (N = 11)	
	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events
Total	31	30 (2.5%)	8	8 (3.6%)	2	1 (1.8%)	1	1 (9.1%)
Sense Lead Revision	16	15 (1.2%)	4	4 (1.8%)	1	1 (1.8%)	1	1 (9.1%)
Stim Lead Revision	5	5 (0.4%)	2	2 (0.9%)	0	0 (0%)	0	0 (0%)
IPG Re-anchored	3	3 (0.2%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
IPG Revision	3	3 (0.2%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Sense Lead Reattached	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Stim Lead Reattached	1	1 (0.1%)	1	1 (0.5%)	1	1 (1.8%)	0	0 (0%)
IPG and Sense Lead Revision	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
System Explant	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Unspecified	0	0 (0%)	1	1 (0.5%)	0	0 (0%)	0	0 (0%)

Note:

Format for Subjects with Events: Number of subjects with at least one event (% of subjects per BMI group)

Subjects could have had more than one follow-up event

Additional Analysis to Address Deficiency Letter January 2023

Implant Adverse Events by Baseline AHI for Patients with a Final Follow-up Visit

Patients that were enrolled in the ADHERE registry by July 22, 2022, had AHI collected at baseline, and had a final follow-up visit with at least one of the following values recorded at the final visit - AHI, ESS, Therapy Usage, CGI, and/or Patient Satisfaction - are included in the following tables regarding implant adverse events.

Implant Adverse Events by Seriousness

Table 15: Implant Adverse Events by Seriousness and Baseline AHI for Patients with a Final Visit

Implant AEs	AHI < / = 65 (N = 1483)		65 < AHI < / = 75 (N = 31)		75 < AHI < / = 100 (N = 26)		AHI > 100 (N = 2)		p-value*
	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events	
Total	67	61 (4.1%)	0	0 (0%)	2	1 (3.8%)	1	1 (50%)	
Serious	9	9 (0.6%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0.1
Non-Serious	58	52 (3.5%)	0	0 (0%)	2	1 (3.8%)	1	1 (50%)	

Note:

Format for Subjects with Events: Number of subjects with at least one event (% of subjects per AHI group).

Subjects could have had more than one implant adverse event.

*p-value derived from Fisher's exact test.

Per the table above, there was no significant difference in the percentage of patients with at least one implant event (serious or non-serious) between the AHI groups for patients with a baseline AHI and final visit (p-value = 0.1).

Serious Implant Adverse Events by Type

Table 16: Serious Implant Adverse Events by Type and Baseline AHI for Patients with a Final Visit

Serious AEs	AHI</=65 (N = 1483)			65<AHI</=75 (N = 31)			75<AHI</=100 (N = 26)			AHI>100 (N = 2)		
	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events
Total	9	9 (0.6%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Revision - Sense Lead	2	2 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Pneumothorax	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Infection	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Hypotension	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Revision - IPG	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Cervical Swelling with submandibular Hematoma	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Revision - Stimulation Lead, IPG	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Intraoperative revision - IPG	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)

Note:

Format for Subjects with Events: Number of subjects with at least one event (% of subjects per AHI group)

Non-Serious Implant Adverse Events by Type

Table 17: Non-Serious Implant Adverse Events by Type and Baseline AHI for Patients with a Final Visit

Non-Serious AEs	AHI</=65 (N = 1483)			65<AHI</=75 (N = 31)			75<AHI</=100 (N = 26)			AHI>100 (N = 2)		
	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events
Total	58	52 (3.5%)	0	0 (0%)	2	1 (3.8%)	1	1 (50%)	1	1 (50%)		
Intraoperative bleeding	8	8 (0.5%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)		
Hematoma	7	7 (0.5%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)		
Neuropraxia	3	3 (0.2%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)		
Unknown	3	3 (0.2%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)		
Incision	3	3 (0.2%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)		
discomfort/irritation												
Speech Difficulties	3	3 (0.2%)	0	0 (0%)	1	1 (3.8%)	0	0 (0%)	0	0 (0%)		
Tongue weakness	3	3 (0.2%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)		
Pneumothorax	2	2 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)		
Lip weakness	2	2 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)		
Postoperative bleeding	2	2 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)		
Urinary retention	2	2 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)		
Facial Swelling	2	2 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)		
Seroma	2	2 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)		
Chest pain	2	2 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)		
Atrial fibrillation	2	2 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)		
Edema	2	2 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)		
Intraoperative nerve repositioning	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)		
Infection	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	1	1 (50%)		
Tongue	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)		
Discomfort/irritation												
Conversion Disorder	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)		
Cervical Swelling	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)		
Nerve weakness	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)		
Technical Challenges	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)	0	0 (0%)		

Table 17: Non-Serious Implant Adverse Events by Type and Baseline AHI for Patients with a Final Visit (*continued*)

Non-Serious AEs	AHI</=65 (N = 1483)		65<AHI</=75 (N = 31)		75<AHI</=100 (N = 26)		AHI>100 (N = 2)	
	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events
Dysphagia	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Wound Dehiscence	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Headache	1	1 (0.1%)	0	0 (0%)	1	1 (3.8%)	0	0 (0%)

Note:

Format for Subjects with Events: Number of subjects with at least one event (% of subjects per AHI group)
Subjects could have had more than one implant adverse event

Revisions - Implant Adverse Events

Table 18: Revisions - Implant Adverse Events by Baseline AHI for Patients with a Final Visit

Revision	AHI</=65 (N = 1483)		65<AHI</=75 (N = 31)		75<AHI</=100 (N = 26)		AHI>100 (N = 2)	
	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events
Total	5	5 (0.3%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Sense Lead	2	2 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
IPG	2	2 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Stimulation Lead, IPG	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)

Note:

Format for Subjects with Events: Number of subjects with at least one event (% of subjects per AHI group)

Implant Adverse Events by Baseline BMI for Patients with a Final Follow-up Visit

Patients that were enrolled in the ADHERE registry by July 22, 2022, had BMI collected at baseline, and had a final follow-up visit with at least one of the following values recorded at the final visit - AHI, ESS, Therapy Usage, CGI, and/or Patient Satisfaction - are included in the following tables regarding implant adverse events.

Implant Adverse Events by Seriousness

Table 19: Implant Adverse Events by Seriousness and Baseline BMI for Patients with a Final Visit

Implant AEs	BMI</=32 (N = 1218)		32<BMI</=35 (N = 222)		35<BMI</=40 (N = 57)		BMI>40 (N = 11)		p-value*
	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events	
Total	55	49 (4%)	10	9 (4.1%)	2	2 (3.5%)	3	3 (27.3%)	
Serious	8	8 (0.7%)	0	0 (0%)	1	1 (1.8%)	0	0 (0%)	3e-02
Non-Serious	47	41 (3.4%)	10	9 (4.1%)	1	1 (1.8%)	3	3 (27.3%)	

Note:

Format for Subjects with Events: Number of subjects with at least one event (% of subjects per BMI group).
Subjects could have had more than one implant adverse event.

*p-value derived from Fisher’s exact test.

Per the table above, at least one BMI group had a significantly different percentage of patients with implant event(s) (serious or non-serious) (p-value < 0.05). However, all adverse events in patients with BMI>40 were non-serious.

Serious Implant Adverse Events by Type

Table 20: Serious Implant Adverse Events by Type and Baseline BMI for Patients with a Final Visit

Serious AEs	BMI < / = 32 (N = 1218)		32 < BMI < / = 35 (N = 222)		35 < BMI < / = 40 (N = 57)		BMI > 40 (N = 11)	
	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events
Total	8	8 (0.7%)	0	0 (0%)	1	1 (1.8%)	0	0 (0%)
Revision - Sense Lead	2	2 (0.2%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Pneumothorax	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Hypotension	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Revision - IPG	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Cervical Swelling with submandibular Hematoma	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Revision - Stimulation Lead, IPG	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Intraoperative revision - IPG	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Infection	0	0 (0%)	0	0 (0%)	1	1 (1.8%)	0	0 (0%)

Note:

Format for Subjects with Events: Number of subjects with at least one event (% of subjects per BMI group)

Non-Serious Implant Adverse Events by Type

Table 21: Non-Serious Implant Adverse Events by Type and Baseline BMI for Patients with a Final Visit

Non-Serious AEs	BMI</=32 (N = 1218)		32<BMI</=35 (N = 222)		35<BMI</=40 (N = 57)		BMI>40 (N = 11)	
	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events
Total	47	41 (3.4%)	10	9 (4.1%)	1	1 (1.8%)	3	3 (27.3%)
Intraoperative bleeding	7	7 (0.6%)	1	1 (0.5%)	0	0 (0%)	0	0 (0%)
Hematoma	5	5 (0.4%)	2	2 (0.9%)	0	0 (0%)	0	0 (0%)
Speech Difficulties	4	4 (0.3%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Neuropraxia	3	3 (0.2%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Incision discomfort/irritation	3	3 (0.2%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Infection	2	2 (0.2%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Pneumothorax	2	2 (0.2%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Unknown	2	2 (0.2%)	1	1 (0.5%)	0	0 (0%)	0	0 (0%)
Postoperative bleeding	2	2 (0.2%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Seroma	2	2 (0.2%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Atrial fibrillation	2	2 (0.2%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Tongue weakness	2	2 (0.2%)	1	1 (0.5%)	0	0 (0%)	0	0 (0%)
Headache	2	2 (0.2%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Lip weakness	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	1	1 (9.1%)
Urinary retention	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	1	1 (9.1%)
Conversion Disorder	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Cervical Swelling	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Facial Swelling	1	1 (0.1%)	1	1 (0.5%)	0	0 (0%)	0	0 (0%)
Nerve weakness	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Technical Challenges	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Chest pain	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	1	1 (9.1%)
Edema	1	1 (0.1%)	0	0 (0%)	1	1 (1.8%)	0	0 (0%)
Intraoperative nerve repositioning	0	0 (0%)	1	1 (0.5%)	0	0 (0%)	0	0 (0%)
Tongue Discomfort/irritation	0	0 (0%)	1	1 (0.5%)	0	0 (0%)	0	0 (0%)

Table 21: Non-Serious Implant Adverse Events by Type and Baseline BMI for Patients with a Final Visit (*continued*)

Non-Serious AEs	BMI < /= 32 (N = 1218)		32 < BMI < /= 35 (N = 222)		35 < BMI < /= 40 (N = 57)		BMI > 40 (N = 11)	
	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events
Dysphagia	0	0 (0%)	1	1 (0.5%)	0	0 (0%)	0	0 (0%)
Wound Dehiscence	0	0 (0%)	1	1 (0.5%)	0	0 (0%)	0	0 (0%)

Note:

Format for Subjects with Events: Number of subjects with at least one event (% of subjects per BMI group)
Subjects could have had more than one implant adverse event

Revisions - Implant Adverse Events

Table 22: Revisions - Implant Adverse Events by Baseline BMI for Patients with a Final Visit

Revision	BMI < /= 32 (N = 1218)		32 < BMI < /= 35 (N = 222)		35 < BMI < /= 40 (N = 57)		BMI > 40 (N = 11)	
	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events	Total Events	Subjects with Events
Total	5	5 (0.4%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Sense Lead	2	2 (0.2%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
IPG	2	2 (0.2%)	0	0 (0%)	0	0 (0%)	0	0 (0%)
Stimulation Lead, IPG	1	1 (0.1%)	0	0 (0%)	0	0 (0%)	0	0 (0%)

Note:

Format for Subjects with Events: Number of subjects with at least one event (% of subjects per BMI group)